

THE ANATOMY OF A PHONE NUMBER

The Architecture of a Phone Number

VOLUME I OF III

Identifiers, numbering plans and 150 years of switching

Why a telephone number is an identifier rather than an address, how the international numbering plan is organised, what the SIM, the handset and the subscription each contribute, and how 150 years of switching technology changed underneath a string of digits that barely changed at all.



A LAYERED STUDY OF TELEPHONE NUMBERING, MOBILE NETWORKS AND PACKET TELEPHONY

PART I / III

About this volume

Why a telephone number is an identifier rather than an address, how the international numbering plan is organised, what the SIM, the handset and the subscription each contribute, and how 150 years of switching technology changed underneath a string of digits that barely changed at all.

This is one of three volumes. Each descends a level further from the digits a human types toward the electromagnetic wave that eventually carries the speech. The volumes are meant to be read in order, but each is self-contained.

VOLUME I

The Architecture of a Phone Number

VOLUME II

Anatomy of a 5G Voice Call

VOLUME III

Packet-Level Telecommunications

Diagrams are numbered by chapter. A list of figures follows the contents, so any diagram can be found directly.

Contents

CH 1	What a Phone Number Actually Is	9
	1.1 The fundamental abstraction	9
	1.2 First distinction: identifier vs address	10
CH 2	The Global Numbering Plan	12
	2.1 The international structure	12
	2.2 Country codes aren't random	13
	2.3 India's number	14
CH 3	The Identity Family: MSISDN, IMSI, IMEI, ICCID	16
	3.1 The phone number's "possessions"	16
	3.2 MSISDN	17
	3.3 IMSI	17
	3.4 IMEI	18

3.5	ICCID	19
CH 4	The Birth of Automatic Switching	21
4.1	Historical architecture: before automatic dialing	21
4.2	The Strowger revolution	22
4.3	The telephone number became an instruction	22
4.4	Rotary dialing	23
CH 5	From Exchanges to National Networks	25
5.1	Local exchanges became networks	25
5.2	Area codes	26
5.3	But numbering isn't necessarily geography anymore	27
5.4	Number portability broke the old model	28
CH 6	Mobility: GSM and the Cellular Model	30
6.1	Mobile networks changed everything	30
6.2	GSM architecture	32
6.3	What happens when someone calls your mobile?	33
6.4	Signaling vs actual voice	34
CH 7	Signalling and the Digital PSTN	36
7.1	SS7	36
7.2	The PSTN became digital	37
7.3	The telephone number survived the transition	38
CH 8	The IP Transition: VoIP, SIP and IMS	40
8.1	VoIP changed the architecture again	40
8.2	SIP	41
8.3	Mobile networks eventually moved toward IMS	42
8.4	The modern stack	43
CH 9	Radio, Mobility Management and Paging	45
9.1	The cellular radio side	45
9.2	Why your phone can ring while moving	46
9.3	Paging	47
CH 10	Databases, Authority and International Routing	48
10.1	Numbering databases	48

10.2 Who controls a phone number?	49
10.3 What happens when you call another country?	49
CH 11 Comparisons and Boundary Cases	51
11.1 The relationship with DNS	51
11.2 Phone number vs IP address	52
11.3 Phone number vs SIM	53
11.4 One person can have multiple numbers	54
CH 12 Special Numbers and Overlaid Services	56
12.1 Emergency numbers	56
12.2 Toll-free numbers	57
12.3 Voicemail	57
12.4 Caller ID	58
12.5 SMS uses the same numbering namespace	59
12.6 WhatsApp demonstrates another layer	60
CH 13 The Whole Evolution in One View	62
13.1 The entire historical evolution	62
13.2 The most important conceptual transformation	64
13.3 The modern conceptual model	66
13.4 The really deep abstraction	68
APP A Glossary of Terms	73

Figures

Fig. 1.1 Structure of +91 98XX YYYYYY	9
Fig. 1.2 Human-readable number → Subscriber database → Your handset	10
Fig. 1.3 Phone number → Location identity → Actual device	11
Fig. 2.1 Structure of + CC N(S)N	12
Fig. 2.2 +91 → 9876543210	13
Fig. 2.3 +91 and its constituent parts	14
Fig. 3.1 Telephony identities → IMSI → IMEI	16
Fig. 3.2 Structure of MSISDN	17
Fig. 3.3 IMSI and its constituent parts	17

Fig. 3.4	SIM/subscription → Telephone number	18
Fig. 3.5	You can move SIM A → to Phone B → IMSI / → same	19
Fig. 3.6	ICCID → MSISDN → mobile equipment	20
Fig. 4.1	Phone A → Phone B	21
Fig. 4.2	Subscriber → Local exchange → Destination subscriber	22
Fig. 4.3	digits → select switching path	23
Fig. 4.4	Digit 2 → select subgroup 7 → select line 4	23
Fig. 4.5	7 → 3 pulses → 5 pulses	24
Fig. 5.1	National network → A1 / a2 / a3 / b1 / b2 / b3	25
Fig. 5.2	subscriber → long-distance network → subscriber	26
Fig. 5.3	Structure of 212 - 555 - 1234	27
Fig. 5.4	number → physical exchange	27
Fig. 5.5	number → current destination	28
Fig. 5.6	Operator A → Operator B	29
Fig. 5.7	Caller → Number analysis → destination network	29
Fig. 6.1	Number → building	30
Fig. 6.2	Number → current location → device	31
Fig. 6.3	Cell A → Cell D	31
Fig. 6.4	PSTN / ISDN / Internet → AuC → SIM	32
Fig. 6.5	Phone → MSC	33
Fig. 6.6	Alice → +91 9876543210	33
Fig. 6.7	Alice's phone → Bob's operator → Bob's phone	34
Fig. 6.8	Signaling → TDM circuits	35
Fig. 6.9	Signaling → RTP over IP	35
Fig. 7.1	Signalling network → Voice network	36
Fig. 7.2	Human voice → sampling → network	37
Fig. 7.3	8 kHz sampling → 64 kbit/s	38
Fig. 7.4	64 kb/s → higher-rate trunk	38
Fig. 7.5	analog → SS7 → cloud-native telecom	39
Fig. 8.1	Telephone → TDM trunk	40
Fig. 8.2	Telephone → SIP infrastructure → destination	41
Fig. 8.3	Caller → Destination	42
Fig. 8.4	PSTN → CSCF / databases → access	43

Fig. 8.5	USER → Signaling / Media → Handset	44
Fig. 9.1	MSISDN → IMEI → actual radio transmission	45
Fig. 9.2	Cell A / Cell B / Cell C → Your telephone number doesn't change	46
Fig. 9.3	MSISDN → Serving network → Current radio access	46
Fig. 9.4	Incoming call → Page devices in relevant area → Connection established	47
Fig. 10.1	Number database → Call routing	48
Fig. 10.2	ITU → Telecom operator → Subscriber	49
Fig. 10.3	Indian phone → US numbering/routing network → At each stage, routing information gets progressively resolved	50
Fig. 11.1	example.com → IP address → server	51
Fig. 11.2	+919876543210 → operator/service → device	52
Fig. 11.3	stable identifier → current network destination	52
Fig. 11.4	Phone number → PDU/session → Internet/service	53
Fig. 11.5	Phone number → SIM/UICC → IMEI	54
Fig. 11.6	Person and its constituent parts	55
Fig. 11.7	One number → multiple endpoints	55
Fig. 11.8	+1 800 xxx xxxx → A B / C	55
Fig. 12.1	Dial emergency number → appropriate emergency center	56
Fig. 12.2	number → destination	57
Fig. 12.3	number → voicemail platform	58
Fig. 12.4	Caller → destination network → display	59
Fig. 12.5	SMS sender → recipient MSISDN → recipient	60
Fig. 12.6	WhatsApp → phone number	61
Fig. 12.7	phone number → application identity → application servers / peers	61
Fig. 13.1	1870s → 1970s-1980s → an identifier above an IP/service infrastructure	63
Fig. 13.2	Physical switching → Digital circuit network → Software-defined telecom	64
Fig. 13.3	NUMBER → Physical circuit	64
Fig. 13.4	NUMBER → Switching path	65
Fig. 13.5	NUMBER → Network destination	65
Fig. 13.6	NUMBER → Current location	65
Fig. 13.7	NUMBER → SERVICE → ENDPOINT	66
Fig. 13.8	HUMAN → Mobility → Imsi/supi /	67
Fig. 13.9	Control plane → Media plane → Audio stream	68
Fig. 13.10	Computer system → IP → machine	69

Fig. 13.11	Telecom system → subscriber identity → device	70
Fig. 13.12	User-facing identity → Location / mobility → IMEI / modem / SIM / radio hardware	71

Introduction

A phone number looks like a tiny string of digits, but underneath it sits on top of roughly 150 years of telecommunications engineering, regulatory allocation, switching theory, signaling protocols, databases, radio networks, and increasingly IP-based routing.

The important conceptual correction is this:

A phone number is not the address of your phone. It is a globally managed identifier that the telephone network uses to discover how to reach your subscription/service.

Your handset has other identities. The SIM has identities. The radio network has identities. The IP network has identities. Your phone number is only one layer of the stack.

Let's peel the whole thing apart.

What a Phone Number Actually Is

1.1 The fundamental abstraction

1.2 First distinction: identifier vs address

1.1 The fundamental abstraction

Suppose your Indian mobile number is:

+91 98XX YYYYYY

At the human level:

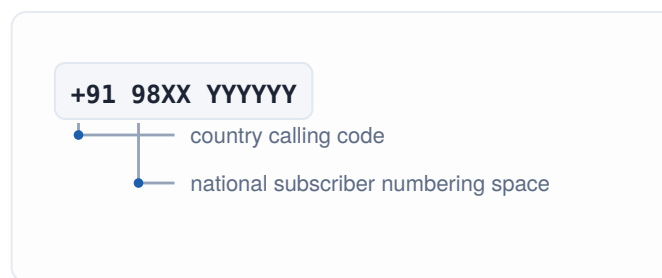


Figure 1.1 Structure of +91 98XX YYYYYY

But the actual telecommunications system does not simply think:

"+9198XXYYYYYY" → radio tower

It is more like:

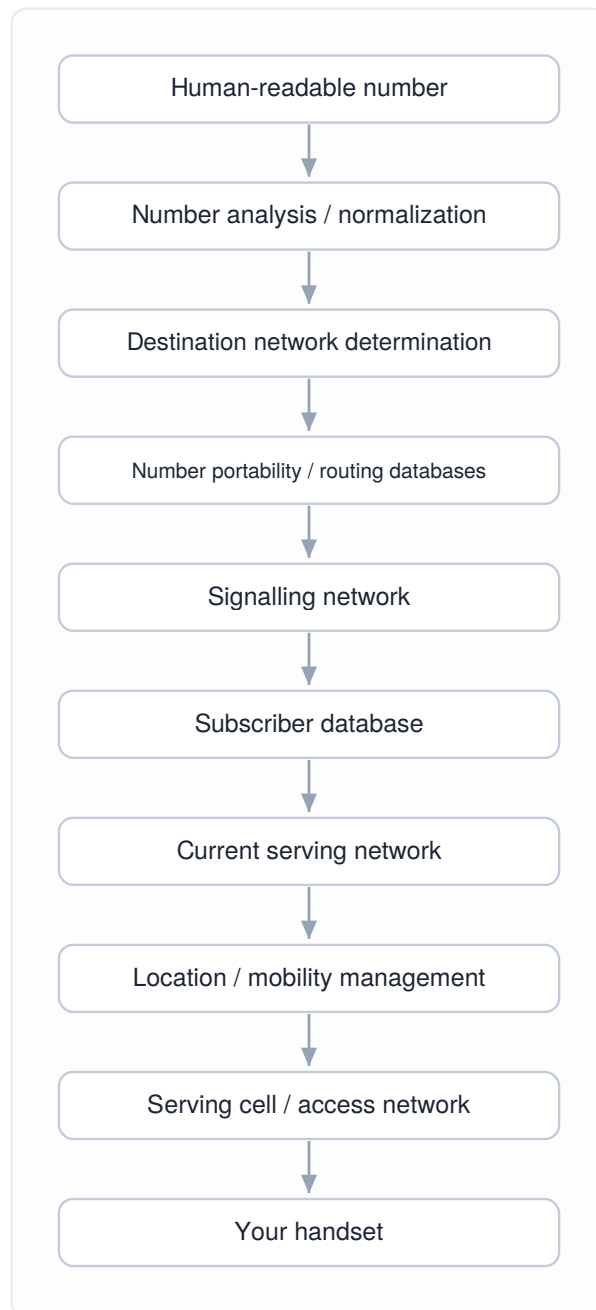


Figure 1.2 Human-readable number → Subscriber database → Your handset

And that architecture has changed dramatically over time.

1.2 First distinction: identifier vs address

This is the most important concept.

In computer networking, you already know:

- MAC address
- IP address
- DNS name
- socket

These aren't the same thing.

Telephony has the same separation.

A simplified model is:

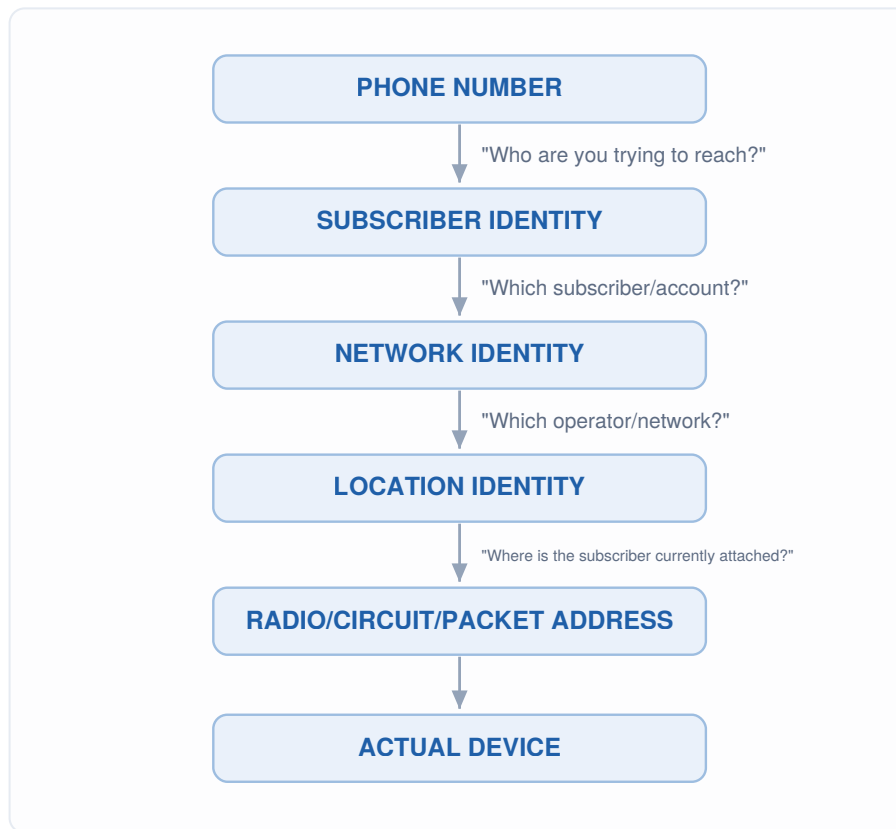


Figure 1.3 Phone number → Location identity → Actual device

So a phone number is closer to a stable logical identifier than a physical network address.

This is why you can change:

- SIM
- handset
- cell tower
- location
- operator

while retaining the same telephone number.

Number portability makes this even more obvious. The FCC describes number portability as allowing customers to keep their number when changing providers, and India implemented nationwide mobile number portability in January 2011. [FCC Complaints](#)

The Global Numbering Plan

2.1 The international structure

2.3 India's number

2.2 Country codes aren't random

2.1 The international structure

The international numbering system is standardized primarily through ITU-T Recommendation E.164.

The current E.164 revision is from February 2026. It defines international public telecommunications numbering structures and currently covers several categories including geographic areas, global services, satellite services, networks, groups of countries, trial resources, and IoT/M2M applications. [ITU](#)

For an ordinary geographic telephone number, conceptually:

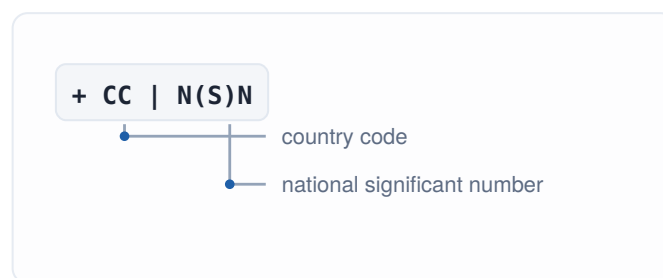


Figure 2.1 Structure of + CC | N(S)N

Where:

CC = Country Code

NSN = National Significant Number

Historically, E.164 has generally operated within a maximum of 15 digits for an international number.

For example:

+91 9876543210

Break it down:

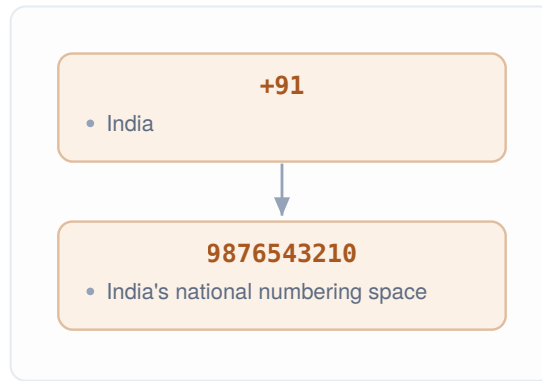


Figure 2.2 +91 → 9876543210

The + isn't actually transmitted as a digit through the telephone network. It is a notation meaning:

"Use the international access procedure appropriate for this country."

For example, depending on where you're dialing from:

+91 9876543210

might be entered using an international dialing prefix such as:

00 91 9876543210

The actual prefix varies by originating country/network.

2.2 Country codes aren't random

The ITU allocates international numbering resources.

Historically, the world was divided into numbering zones.

Very roughly:

- Zone 1** → North America
- Zone 2** → Africa
- Zone 3/4** → Europe
- Zone 5** → Latin America
- Zone 6** → Southeast Asia/Oceania
- Zone 7** → Russia/Kazakhstan
- Zone 8** → East Asia
- Zone 9** → West/South/Central Asia + Middle East

Hence:

- India +91
- Pakistan +92
- Afghanistan +93
- Sri Lanka +94

The exact allocation architecture is more complicated than simply "zone + country", because numbering resources evolved historically.

The ITU maintains the authoritative international allocation framework. ^{ITU}

2.3 India's number

For an ordinary Indian mobile number:

+91 XXXXXXXXXX

you normally have:

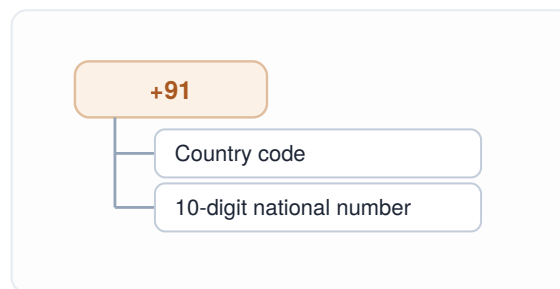


Figure 2.3 +91 and its constituent parts

Example:

+91 9876543210

Historically, Indian mobile numbering ranges were strongly associated with:

- operator
- service area/circle
- subscriber range

But number portability destroyed the simplistic assumption that a prefix tells you the current operator.

For example, you could originally receive:

98XX → Operator A

and later port:

98XX → Operator B

while retaining:

98XX YYYYYY

India's MNP framework was introduced initially in Haryana as a pilot in November 2010 and implemented across all telecom service areas in January 2011. (Press Information Bureau)

That little ten-digit number therefore became considerably less informative than people assume.

The Identity Family: MSISDN, IMSI, IMEI, ICCID

3.1 The phone number's "possessions"

3.2 MSISDN

3.3 IMSI

3.4 IMEI

3.5 ICCID

3.1 The phone number's "possessions"

If you're asking what identities exist around a mobile number, this is where things become interesting.

A modern mobile subscription can involve something like:

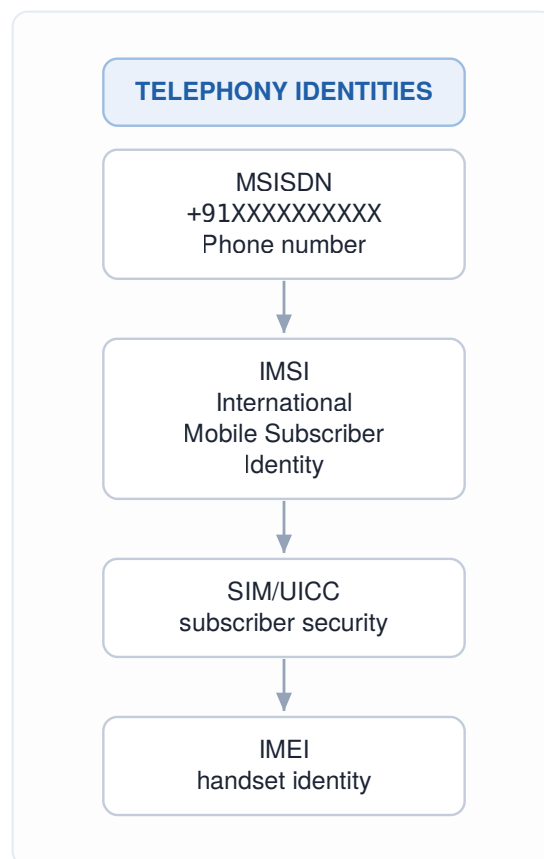


Figure 3.1 Telephony identities → IMSI → IMEI

These are not interchangeable.

3.2 MSISDN

The telephone number associated with a mobile subscriber is generally called the:

MSISDN

Mobile Station ISDN Number.

GSMA defines MSISDN as the structured identifier for the subscriber indicating the A-number. GSMA

Conceptually:

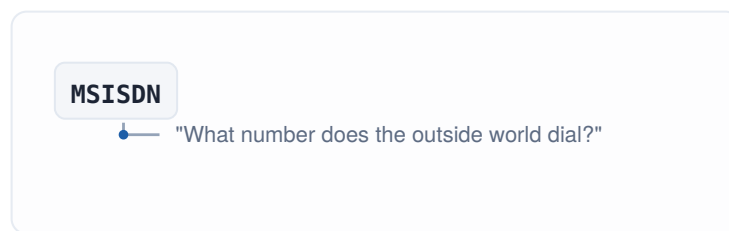


Figure 3.2 Structure of MSISDN

Example:

+919876543210

That's the number your friend knows.

3.3 IMSI

Now we get deeper.

The SIM/subscriber has an:

IMSI

International Mobile Subscriber Identity.

Conceptually:

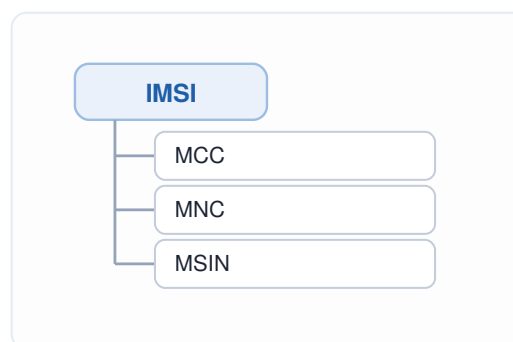


Figure 3.3 IMSI and its constituent parts

Typically:

MCC = Mobile Country Code

MNC = Mobile Network Code

MSIN = Mobile Subscriber Identification Number

The IMSI identifies the subscription within the mobile system.

It is not the same thing as your phone number.

This distinction is fundamental:

- | | |
|--------------------|--------------------------------|
| • MSISDN | • IMSI |
| • telephone number | • mobile subscription identity |

3GPP/ETSI specifications explicitly maintain subscriber information such as IMSI, MSISDN, IMEI and related mobility/security parameters in mobile-network databases. (ETSI)

3.4 IMEI

Then there's the:

IMEI

International Mobile Equipment Identity.

This identifies the equipment, not the subscriber.

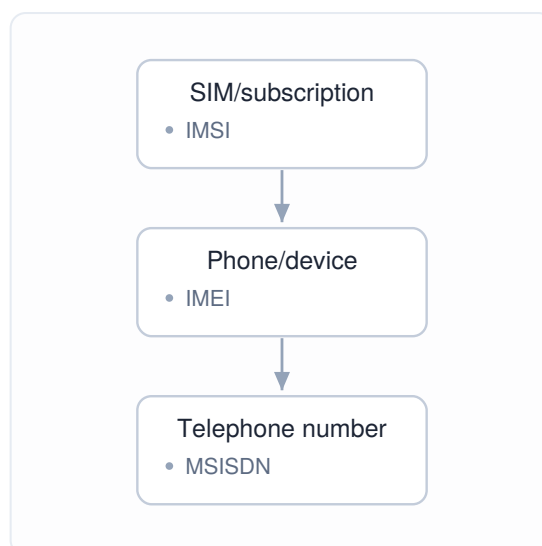


Figure 3.4 SIM/subscription → Telephone number

Therefore:

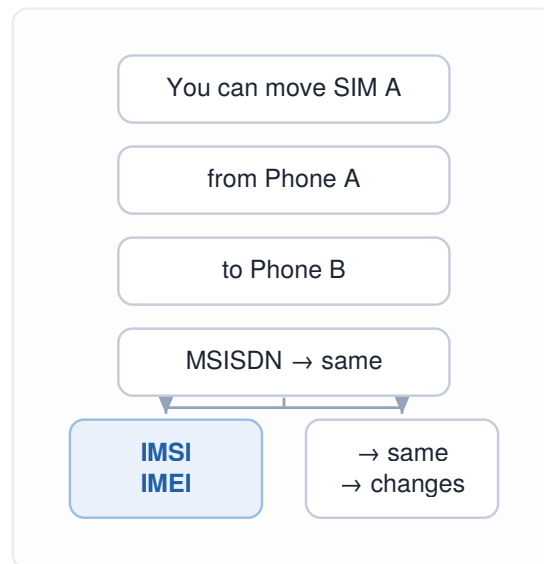


Figure 3.5 You can move SIM A → to Phone B → IMSI / → same

That is the reason the network can distinguish:

"This subscriber"

from:

"This physical handset."

3.5 ICCID

The physical SIM/UICC also has an:

ICCID

Integrated Circuit Card Identifier.

Think:

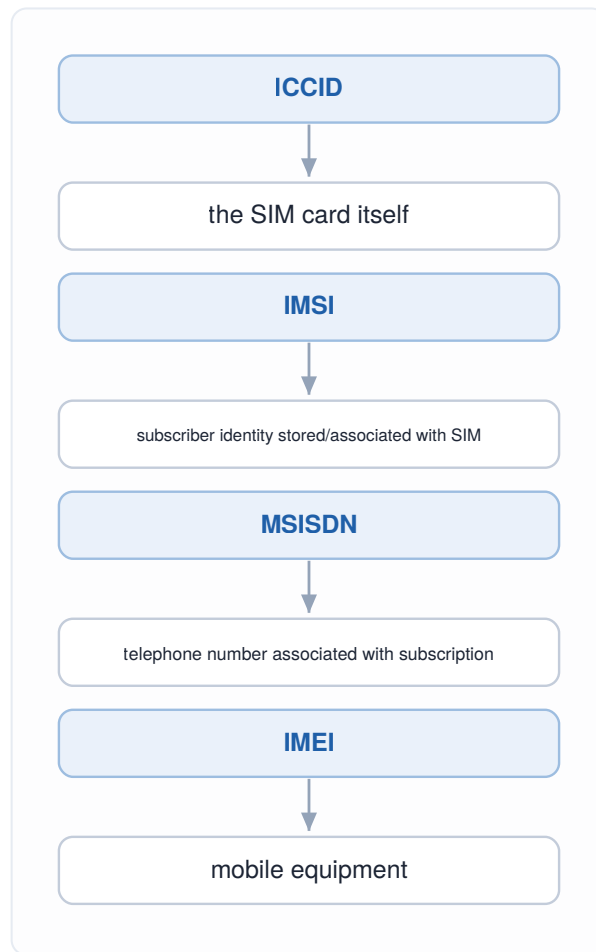


Figure 3.6 ICCID → MSISDN → mobile equipment

Modern eSIMs complicate the physical-card intuition, because the subscription credentials can be provisioned into an embedded secure element rather than a removable plastic SIM.

Human beings invented a card, then removed the card, then embedded the card. Progress, apparently, is making the card invisible.

The Birth of Automatic Switching

4.1 Historical architecture: before automatic dialing

4.2 The Strowger revolution

4.3 The telephone number became an instruction

4.4 Rotary dialing

4.1 Historical architecture: before automatic dialing

Now let's go back to the nineteenth century.

Early telephone networks were extremely primitive by modern standards.

Imagine:

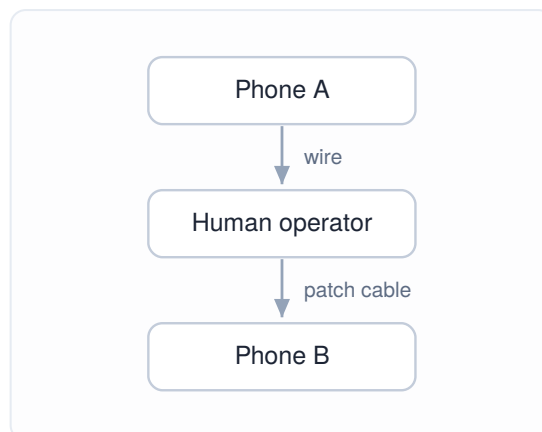


Figure 4.1 Phone A → Phone B

You picked up your telephone and effectively said:

"Connect me to John."

The operator manually connected circuits.

There wasn't necessarily a globally meaningful telephone number in the modern sense.

The network could be organized around:

- subscriber name
- exchange
- local identifier

Telephone directories subsequently became an essential component of this social/network infrastructure. The Library of Congress has surviving telephone directories from the 1890s onward, showing how exchange-based addressing became institutionalized. (Research Guides)

4.2 The Strowger revolution

Then came automatic switching.

Almon Strowger developed an automatic telephone exchange in the late nineteenth century.

The famous motivation is almost absurdly human: Strowger, an undertaker, believed a competing undertaker's wife, who worked as a telephone operator, was redirecting his customers' calls.

So he developed an automated switching system that didn't require her.

The first commercial Strowger exchange began operating in the early 1890s. (IEEE Spectrum)

Architecture:

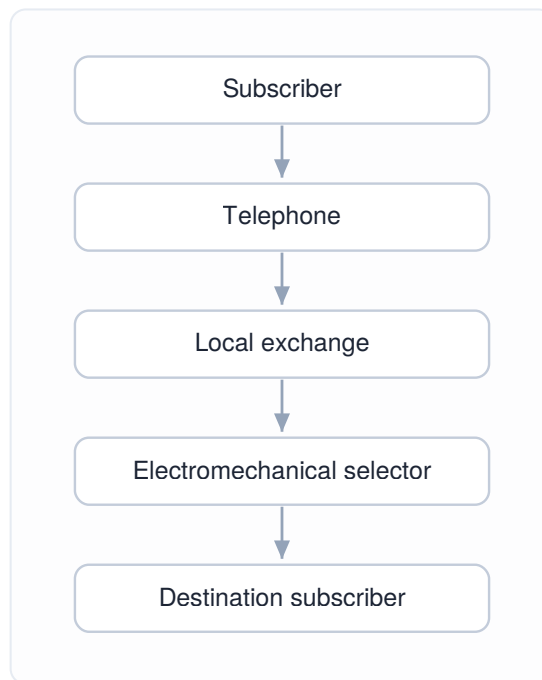


Figure 4.2 Subscriber → Local exchange → Destination subscriber

Now the caller could directly select a destination.

That fundamentally changed what a number meant.

4.3 The telephone number became an instruction

This is a subtle but important idea.

A telephone number isn't merely:

ID = 123456

It was historically closer to:

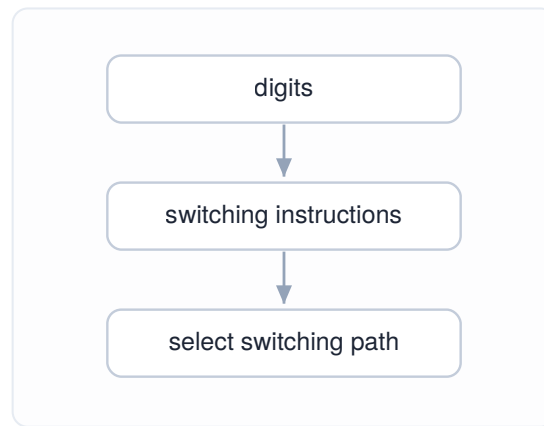


Figure 4.3 digits → select switching path

For example, imagine a primitive exchange:

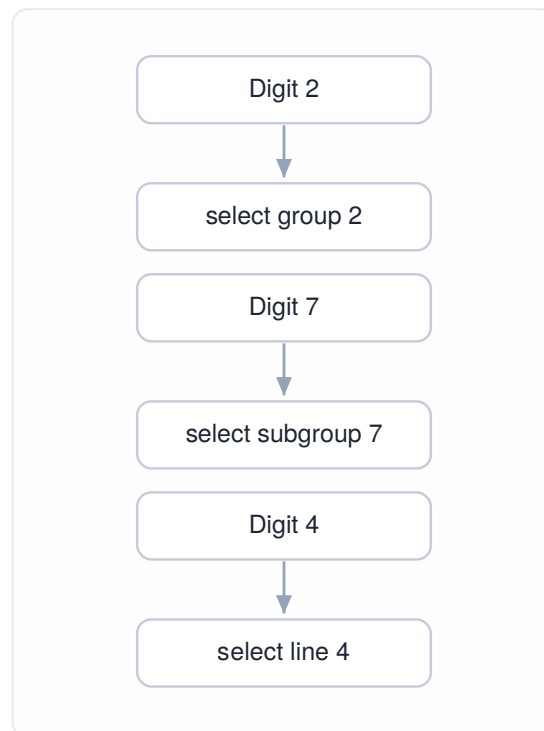


Figure 4.4 Digit 2 → select subgroup 7 → select line 4

The digits were progressively interpreted by switching equipment.

This is digit analysis.

Modern networks still do digit analysis, although the machinery is now software, signaling systems and databases instead of giant electromechanical selectors.

4.4 Rotary dialing

Then came the rotary dial.

The user physically generated pulses corresponding to digits.

Conceptually:

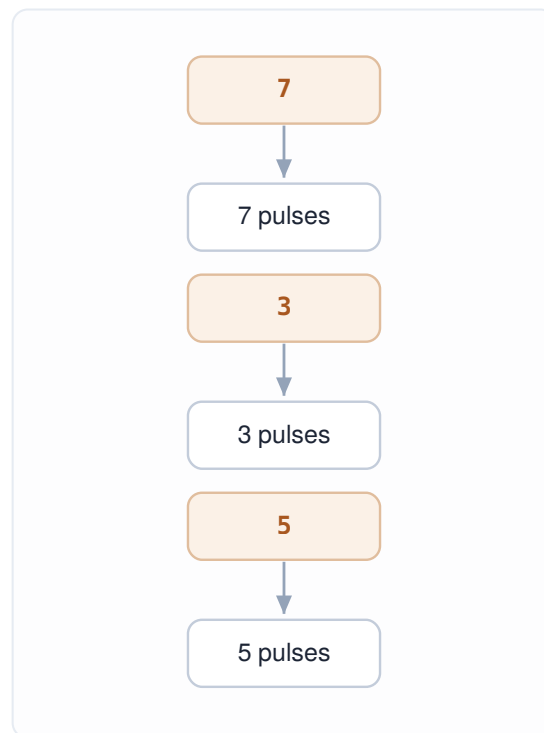


Figure 4.5 7 → 3 pulses → 5 pulses

The exchange counted them.

Later, DTMF replaced pulse dialing for many applications.

DTMF gives us the familiar telephone keypad:

- 1 2 3
- 4 5 6
- 7 8 9
- * 0 #

Each key generates two simultaneous frequencies.

For example:

5
770 Hz + 1336 Hz

The receiver detects the pair.

This became enormously useful for:

- automated menus
- IVR
- banking systems
- voicemail
- remote control systems

From Exchanges to National Networks

5.1 Local exchanges became networks

5.2 Area codes

5.3 But numbering isn't necessarily geography anymore

5.4 Number portability broke the old model

5.1 Local exchanges became networks

One exchange isn't enough for a city.

So the architecture became hierarchical:

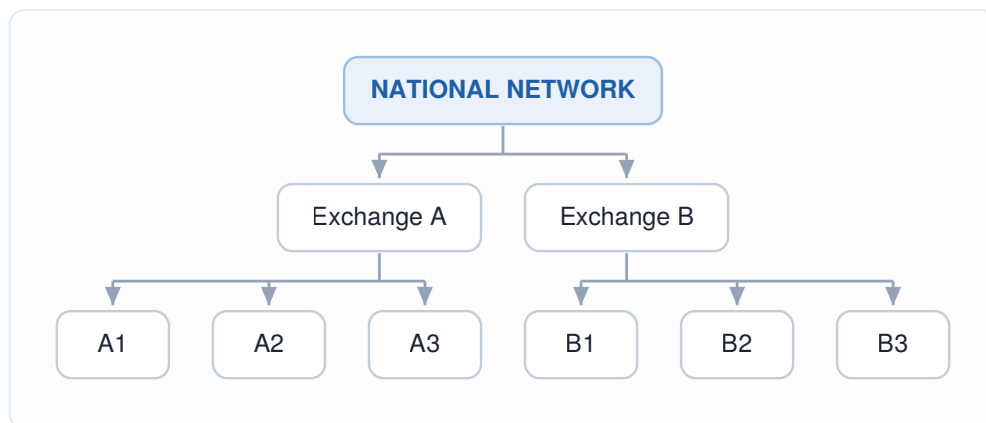


Figure 5.1 National network → A1 / a2 / a3 / b1 / b2 / b3

Eventually:

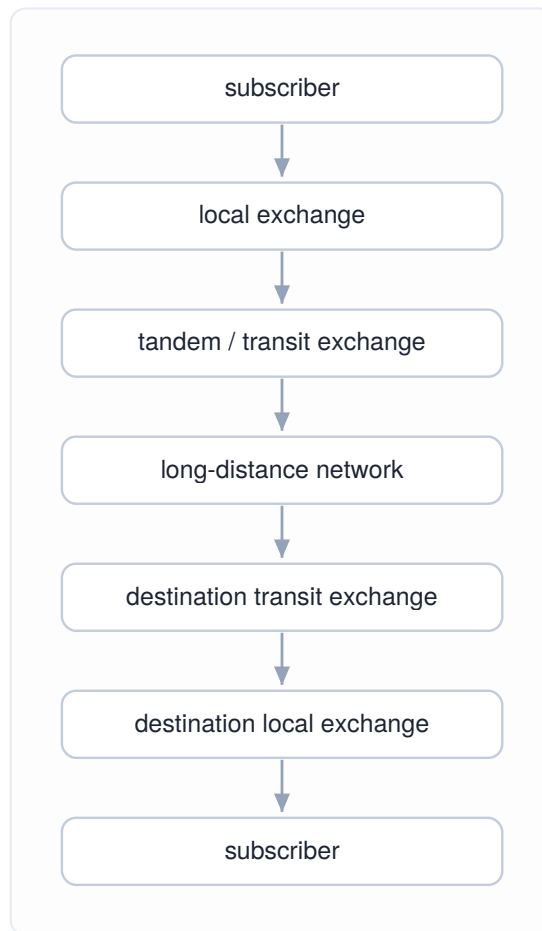


Figure 5.2 subscriber → long-distance network → subscriber

The number structure evolved partly because the network needed enough information to progressively determine where to send the call.

5.2 Area codes

The United States and Canada provide a particularly famous example.

AT&T developed the North American Numbering Plan (NANP) in 1947 to facilitate direct long-distance dialing, with implementation beginning in 1951. (Nanpa)

The basic structure became:

NPA - NXX - XXXX

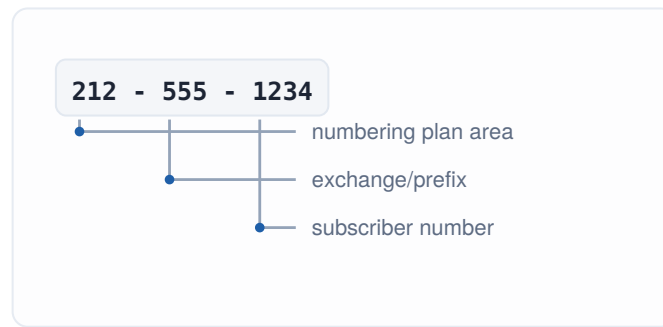


Figure 5.3 Structure of 212 - 555 - 1234

Internationally:

+1 212 555 1234

The 1 identifies the NANP world.

The area code helped the switching system determine the broad destination.

5.3 But numbering isn't necessarily geography anymore

This is another major evolutionary transition.

Originally:

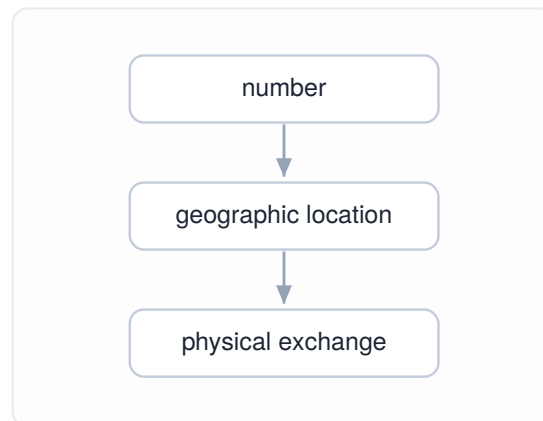


Figure 5.4 number → physical exchange

Modern systems increasingly became:

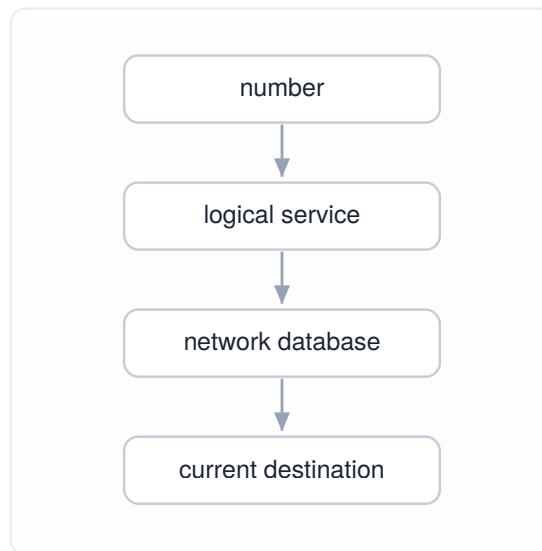


Figure 5.5 number → current destination

This enables:

- mobile numbers
- toll-free numbers
- VoIP
- virtual numbers
- corporate numbering
- unified communications
- number portability
- cloud telephony

The ITU's E.164 framework explicitly covers not only geographic numbering but global services, networks, satellite services and newer IoT/M2M numbering categories. ^{ITU}

5.4 Number portability broke the old model

Suppose:

+91 9876543210

was originally allocated to:

Operator A

You port it:

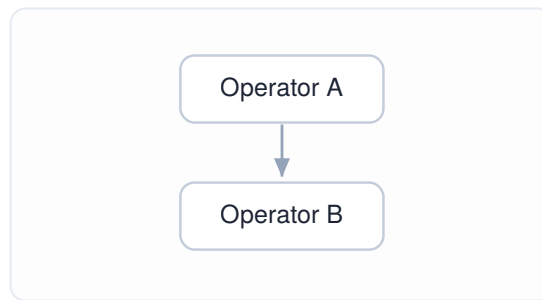


Figure 5.6 Operator A → Operator B

The number doesn't change.

Therefore:

PREFIX

CURRENT OPERATOR

The network needs additional routing intelligence.

A simplified architecture:

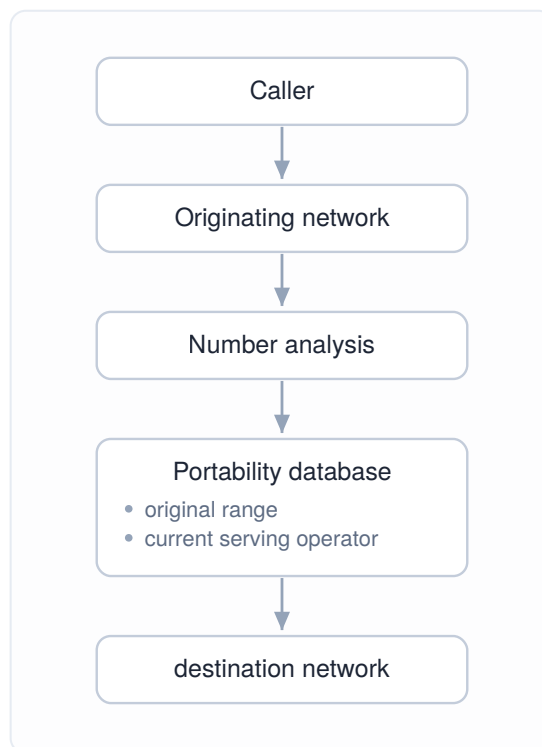


Figure 5.7 Caller → Number analysis → destination network

The FCC's 1996 number portability order explicitly discusses database architecture and routing, illustrating how portability transformed numbering from a purely hierarchical routing mechanism into a database-assisted system. (FCC Documentation)

Mobility: GSM and the Cellular Model

6.1 Mobile networks changed everything

6.2 GSM architecture

6.3 What happens when someone calls your mobile?

6.4 Signaling vs actual voice

6.1 Mobile networks changed everything

Now introduce mobility.

A landline is approximately:

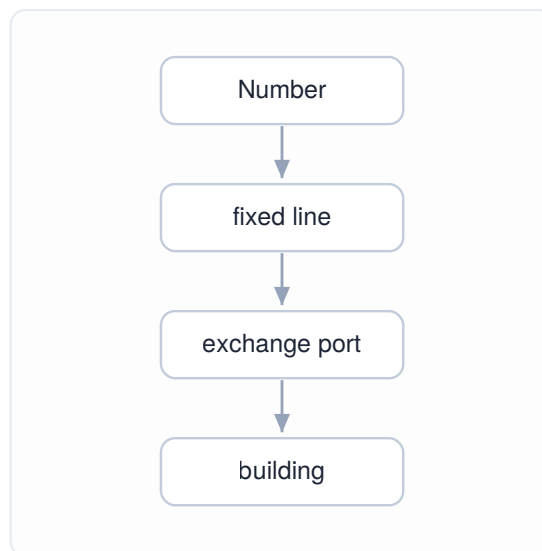


Figure 6.1 Number → building

A mobile phone is:

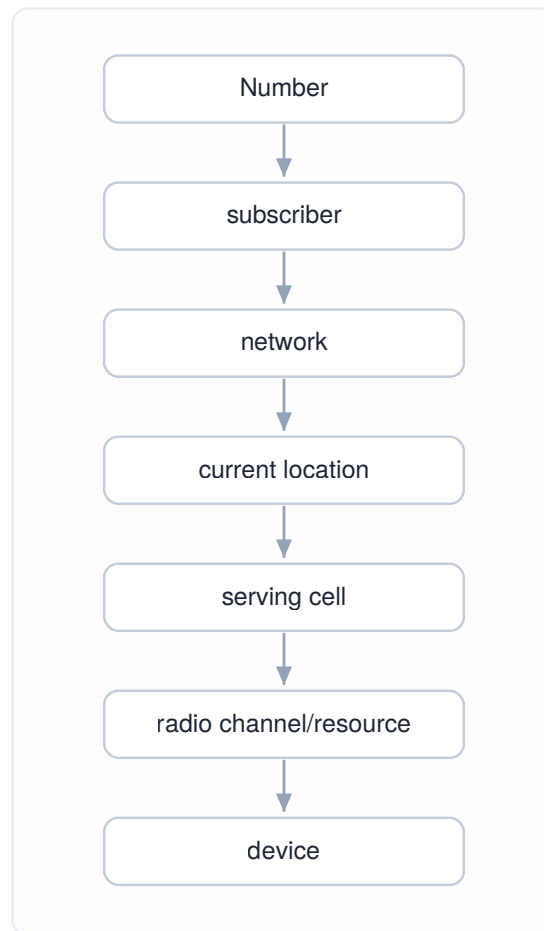


Figure 6.2 Number → current location → device

The subscriber can move:

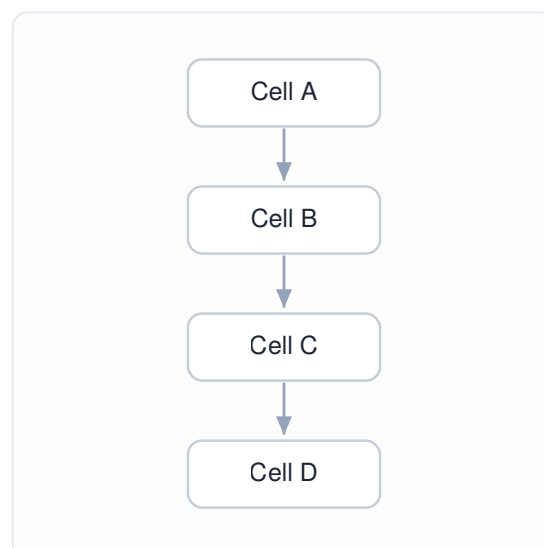


Figure 6.3 Cell A → Cell D

while:

MSISDN = unchanged

IMSI = unchanged

The network therefore needs mobility management.

6.2 GSM architecture

The classic GSM architecture looked approximately like:

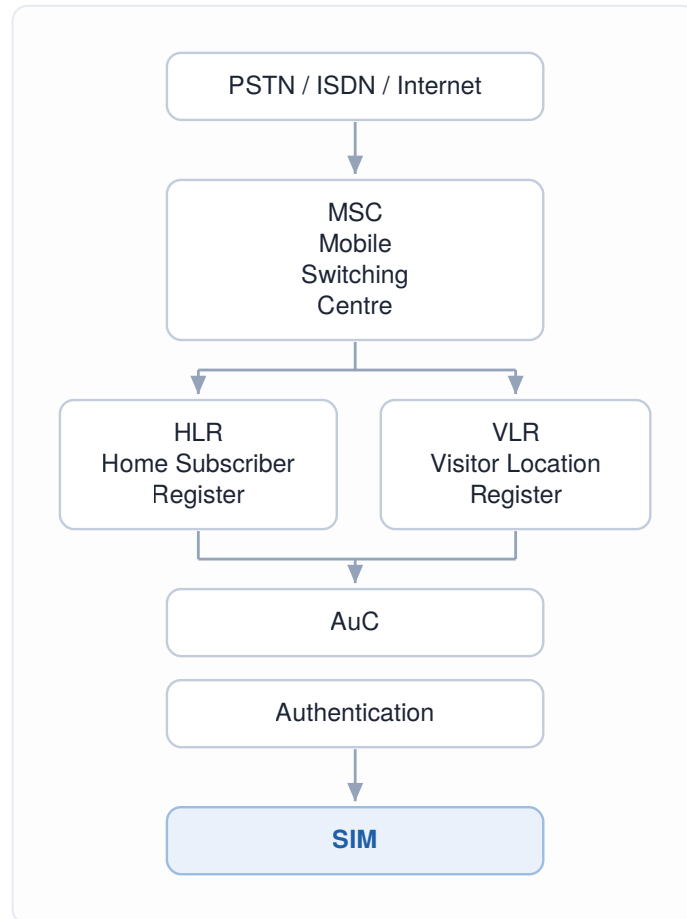


Figure 6.4 PSTN / ISDN / Internet → AuC → SIM

And the radio side:



Figure 6.5 Phone → MSC

So a GSM phone number was only the entry point.

The network then used subscriber and mobility databases to determine where the subscriber currently was.

3GPP/ETSI specifications document subscriber parameters such as IMSI, MSISDN, HLR/VLR information, IMEI and authentication material. (ETSI)

6.3 What happens when someone calls your mobile?

Suppose:



Figure 6.6 Alice → +91 9876543210

A simplified modern cellular process is:

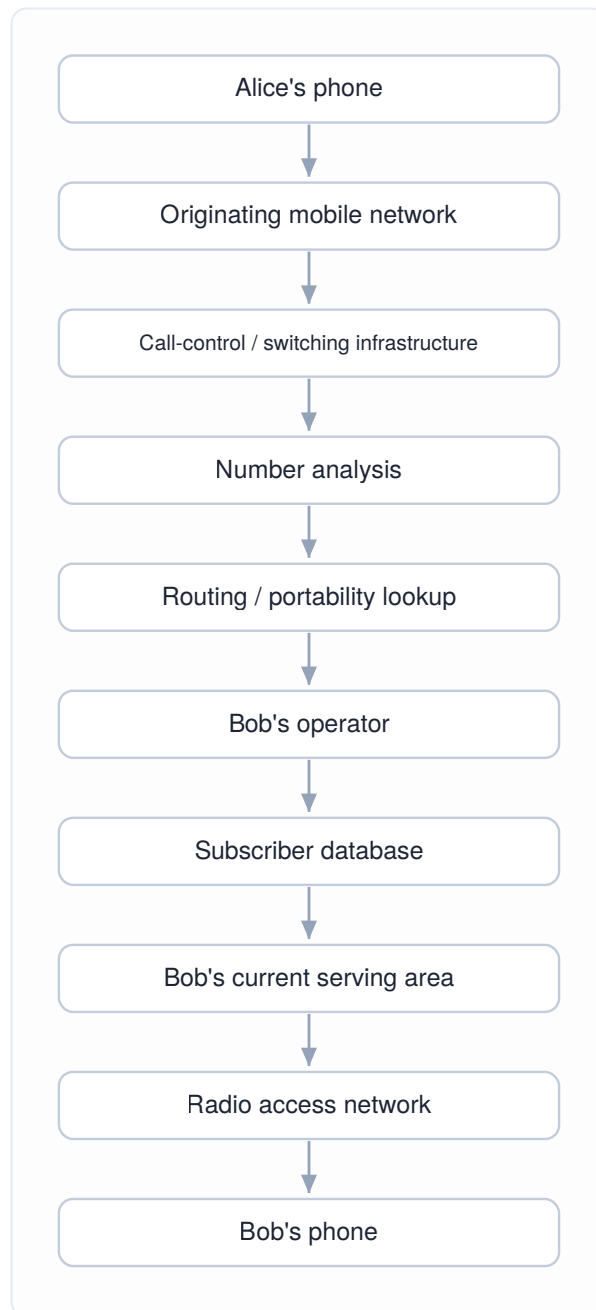


Figure 6.7 Alice's phone → Bob's operator → Bob's phone

The network then establishes the media path.

For traditional circuit-switched voice:

signalling path

voice path

This distinction becomes critical.

6.4 Signaling vs actual voice

This is one of the most important concepts in telephony.

There are essentially two things happening:

- CONTROL PLANE
- "What should happen?"
- DATA/MEDIA PLANE
- "Carry the actual conversation."

Historically:

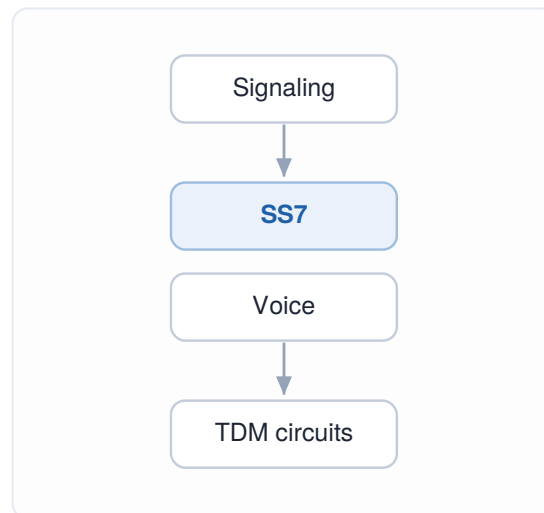


Figure 6.8 Signaling → TDM circuits

Modern systems:

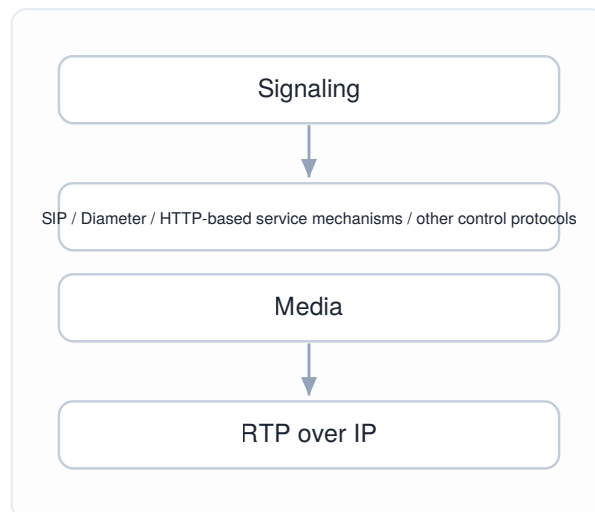


Figure 6.9 Signaling → RTP over IP

So when you dial:

+91 9876543210

the digits don't become the audio stream.

They trigger a chain of control-plane operations that ultimately establish a media session.

Signalling and the Digital PSTN

7.1 SS7

7.3 The telephone number survived the transition

7.2 The PSTN became digital

7.1 SS7

The old PSTN became vastly more sophisticated with digital signaling.

One of the central technologies was:

SS7 — Signaling System No. 7

Instead of using the same voice channel to perform all signaling, networks could use dedicated signaling infrastructure.

Conceptually:

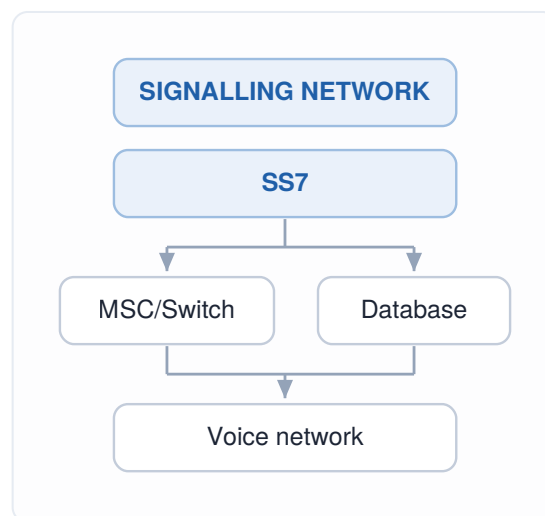


Figure 7.1 Signalling network → Voice network

This allowed things like:

- call setup
- call release
- caller identification
- roaming
- database queries
- number translation

- mobility management
- billing-related signaling

The evolution from electromechanical switching toward electronic switching and SS7 is one of the major transitions in telephone-network architecture. (IEEE Spectrum)

7.2 The PSTN became digital

The old telephone system originally transmitted analog voice.

Then came digital voice.

The basic idea:

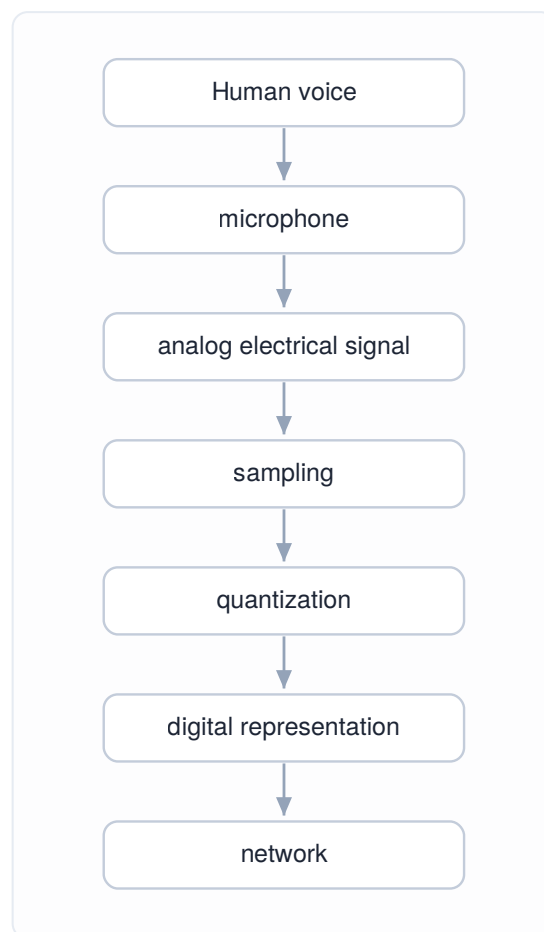


Figure 7.2 Human voice → sampling → network

Classic telephone PCM uses:



Figure 7.3 8 kHz sampling → 64 kbit/s

Hence the famous:

DS0 = 64 kbit/s

Voice channels could then be multiplexed:

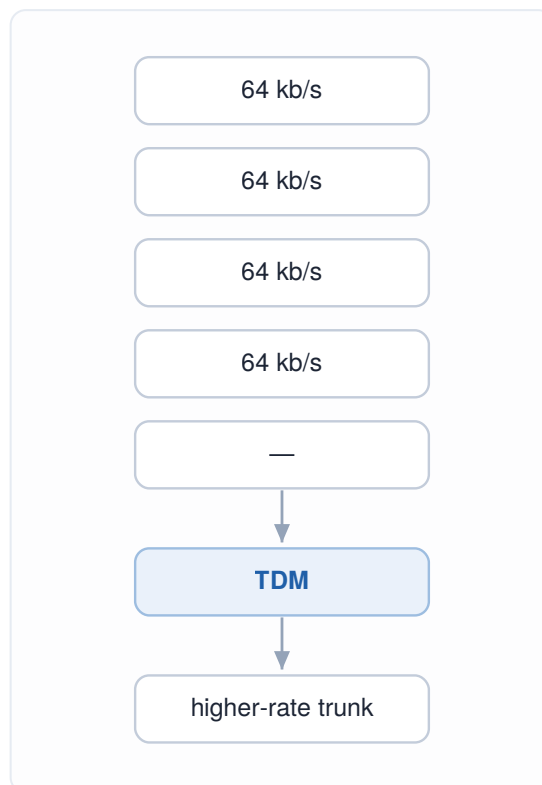


Figure 7.4 64 kb/s → higher-rate trunk

This massively improved network efficiency.

7.3 The telephone number survived the transition

Here's the beautiful engineering hack.

The underlying network could change:

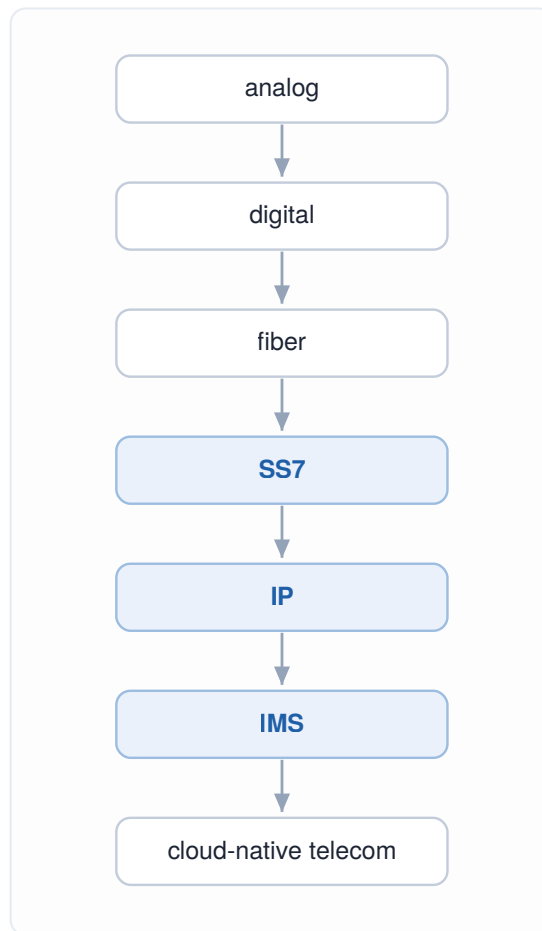


Figure 7.5 analog → SS7 → cloud-native telecom

while the user-facing number remained:

+91 9876543210

That is why numbering systems are so valuable.

They provide application-level stability despite infrastructure-level evolution.

It's analogous to:

DNS name

surviving:

- server migration
- load balancer change
- IP change
- datacenter migration

The identifier persists while the underlying route changes.

The IP Transition: VoIP, SIP and IMS

8.1 VoIP changed the architecture again

8.2 SIP

8.3 Mobile networks eventually moved toward IMS

8.4 The modern stack

8.1 VoIP changed the architecture again

Voice eventually moved onto IP networks.

Instead of:

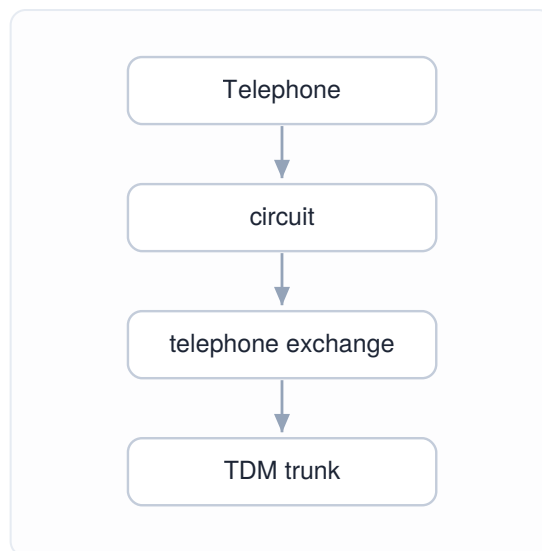


Figure 8.1 Telephone → TDM trunk

we can have:

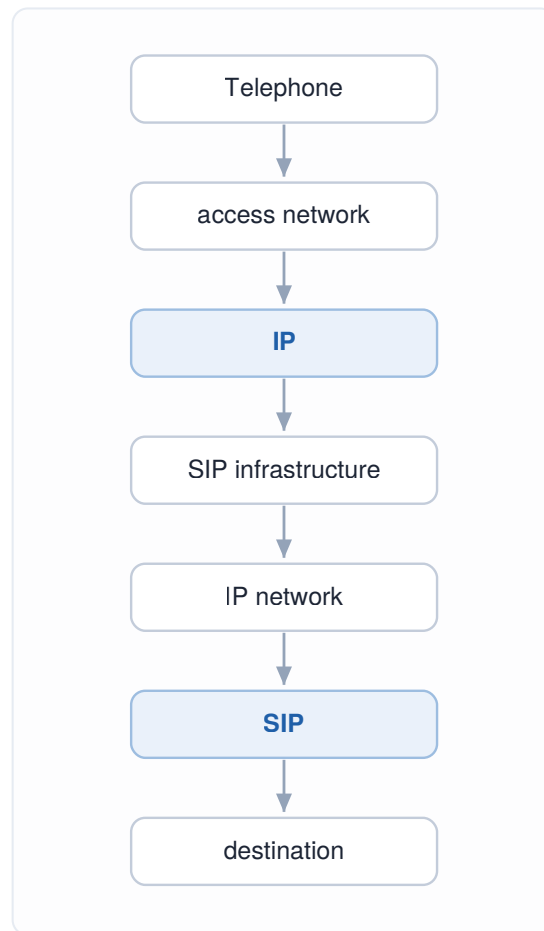


Figure 8.2 Telephone → SIP infrastructure → destination

Now the telephone number can be mapped into an IP-oriented architecture.

This produced:

- VoIP
- SIP phones
- softphones
- hosted PBXs
- cloud telephony
- virtual numbers

8.2 SIP

Modern IP telephony commonly uses SIP, Session Initiation Protocol, for session signaling.

Conceptually:

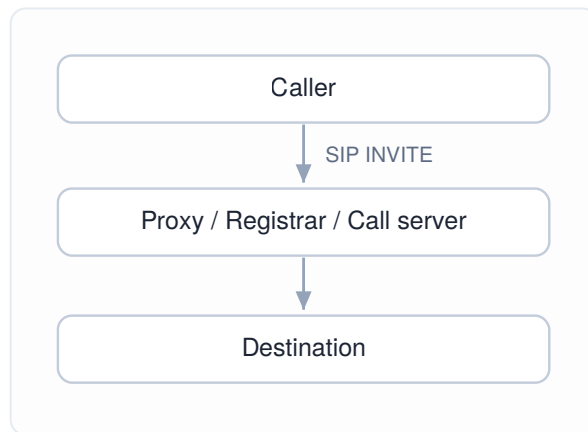


Figure 8.3 Caller → Destination

The actual audio can then be carried using:

RTP

- SIP
- "Let's establish a call."
- RTP
- "Here is the actual audio."

Again:

control plane ≠ media plane

This is the same conceptual separation you see in distributed computing everywhere.

8.3 Mobile networks eventually moved toward IMS

Modern LTE/5G voice architecture is much more IP-centric.

A simplified conceptual architecture:

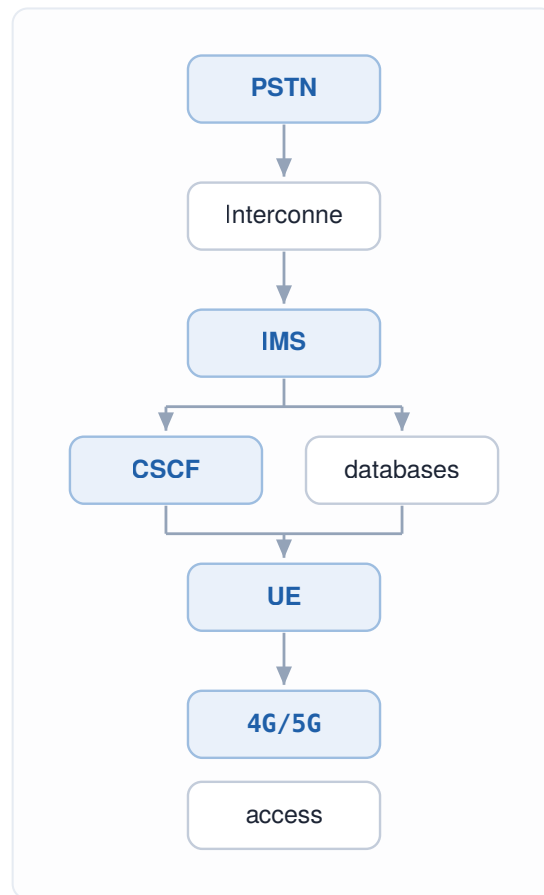


Figure 8.4 PSTN → CSCF / databases → access

IMS means:

IP Multimedia Subsystem

It provides service/control architecture for IP-based telecommunications.

In LTE, voice can be provided through:

VoLTE

In 5G:

VoNR

depending on deployment.

8.4 The modern stack

If you make the abstraction sufficiently deep, a modern phone call can be thought of as:

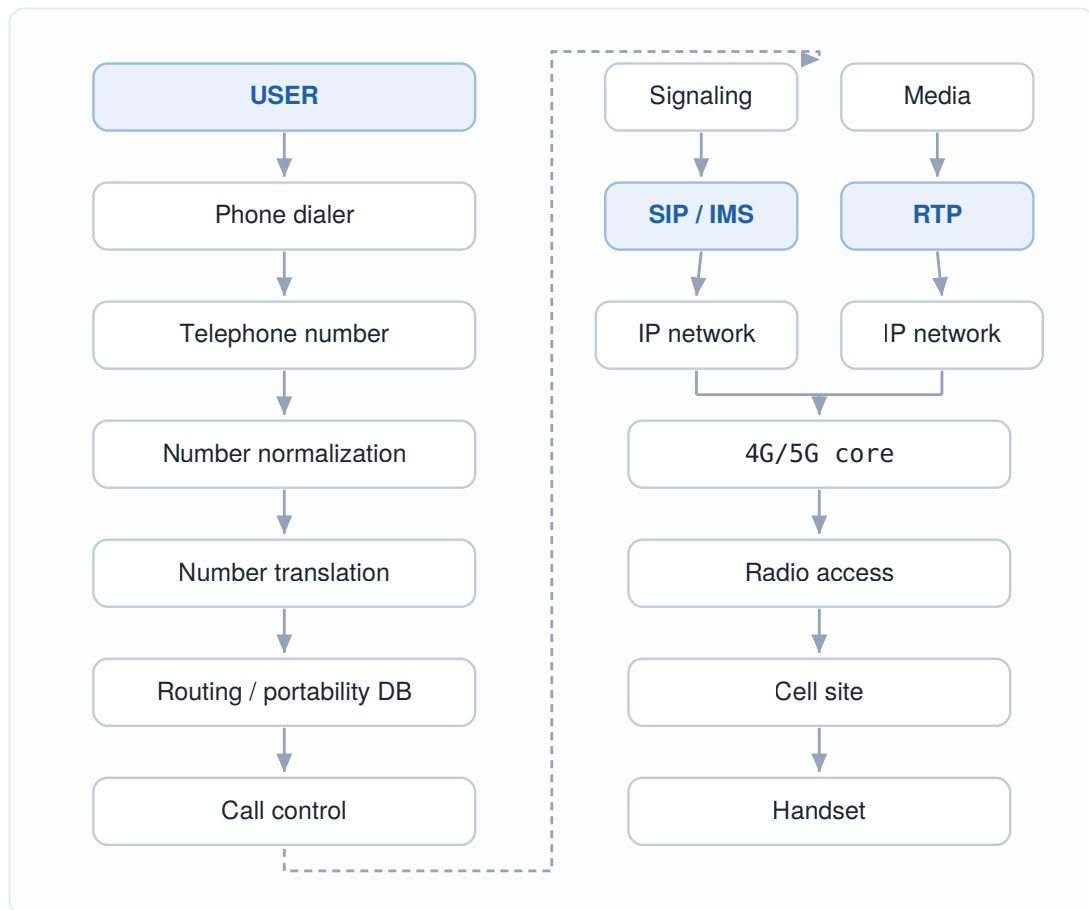


Figure 8.5 USER → Signaling / Media → Handset

That's the conceptual modern architecture.

Radio, Mobility Management and Paging

9.1 The cellular radio side

9.3 Paging

9.2 Why your phone can ring while moving

9.1 The cellular radio side

The telephone number doesn't directly identify:

- tower
- resource block
- frequency
- antenna

Those are dynamically managed.

A modern 5G phone may have:

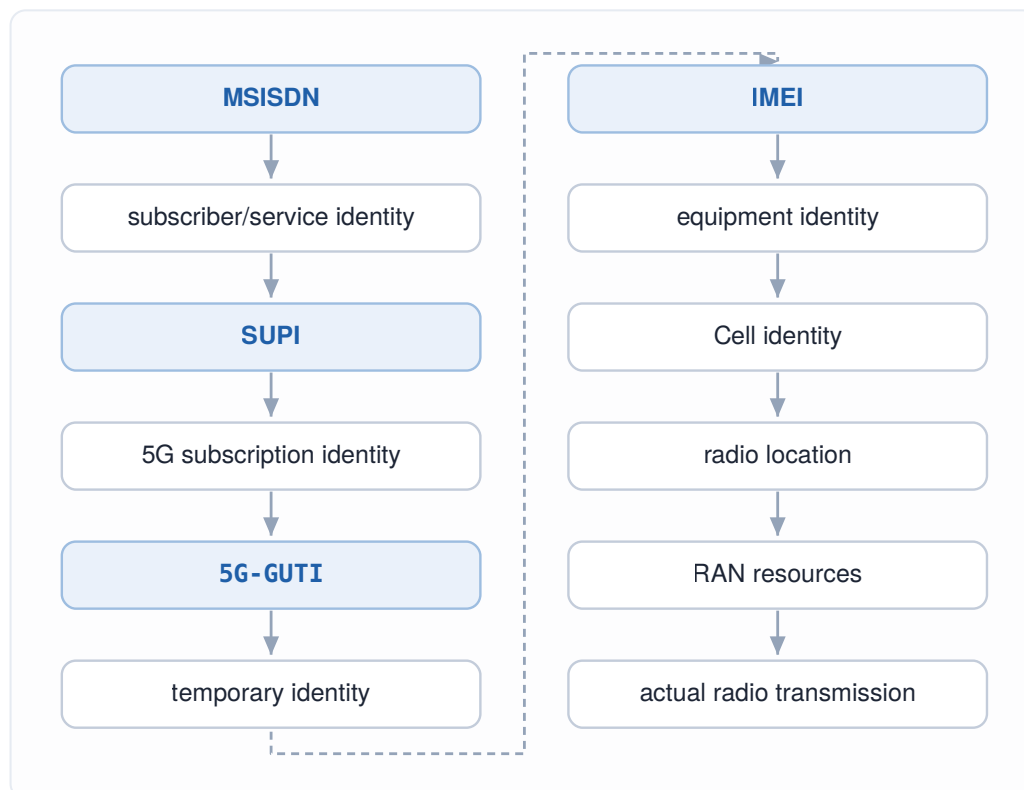


Figure 9.1 MSISDN → IMEI → actual radio transmission

The network deliberately uses temporary identifiers where possible rather than constantly broadcasting permanent subscriber identities.

9.2 Why your phone can ring while moving

Imagine:

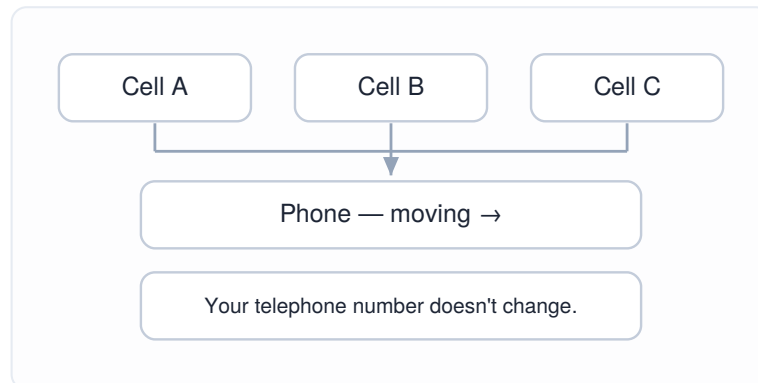


Figure 9.2 Cell A / Cell B / Cell C → Your telephone number doesn't change

Instead, mobility management updates the network's understanding of where your subscription can currently be reached.

Simplified:

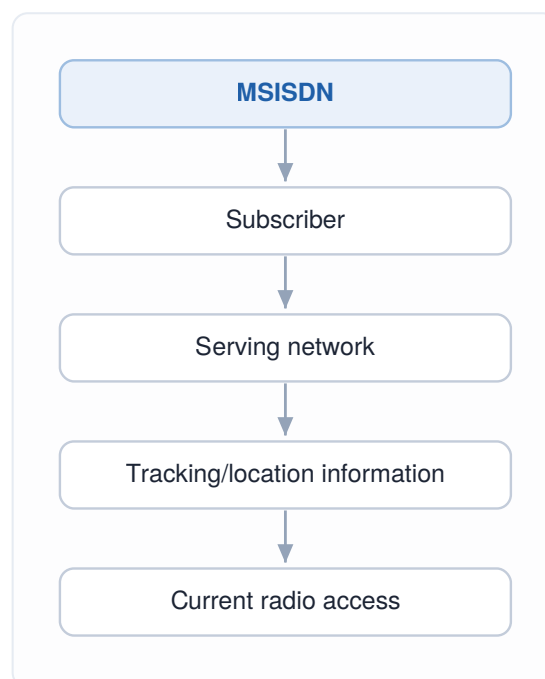


Figure 9.3 MSISDN → Serving network → Current radio access

During mobility, the network performs procedures such as:

- registration
- location updates
- handover
- paging
- authentication
- security context management

So your number is stable while the network attachment state is dynamic.

9.3 Paging

Suppose your phone is idle.

The network doesn't continuously transmit:

"WHERE IS +91 9876543210?"

to every tower on Earth.

That would be an impressively stupid use of infrastructure.

Instead, the network maintains location information at an appropriate granularity.

When a call arrives:

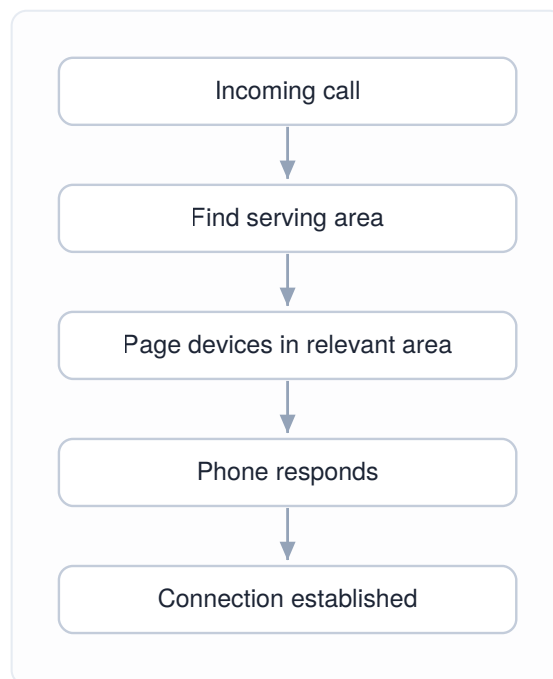


Figure 9.4 Incoming call → Page devices in relevant area → Connection established

This is one reason mobility management is such a fundamental part of cellular architecture.

Databases, Authority and International Routing

10.1 Numbering databases

10.2 Who controls a phone number?

10.3 What happens when you call another country?

10.1 Numbering databases

Modern telecommunications heavily depends on databases.

Conceptually:

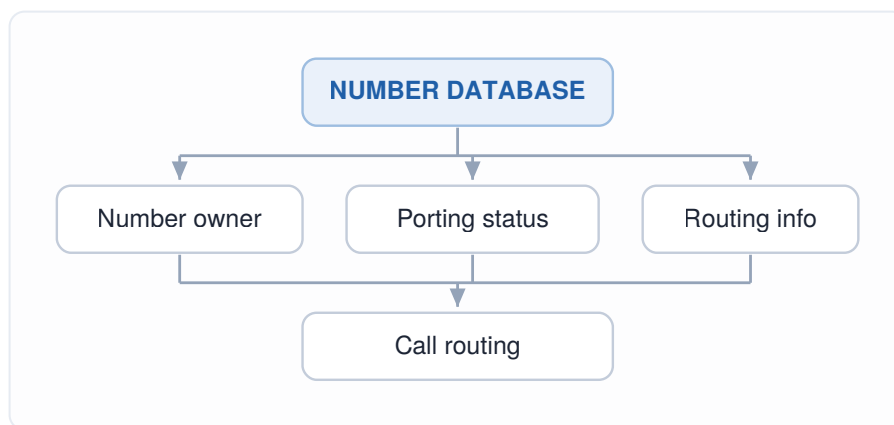


Figure 10.1 Number database → Call routing

The exact databases and protocols vary enormously between countries and technologies.

There is no single magical:

GLOBAL_PHONE_NUMBER_DATABASE

Instead there are:

- international numbering administration
- national numbering authorities
- operator databases
- portability databases
- subscriber databases
- routing databases
- service databases

This is a distributed administrative and technical ecosystem.

10.2 Who controls a phone number?

This is another layer people often misunderstand.

There are several levels:

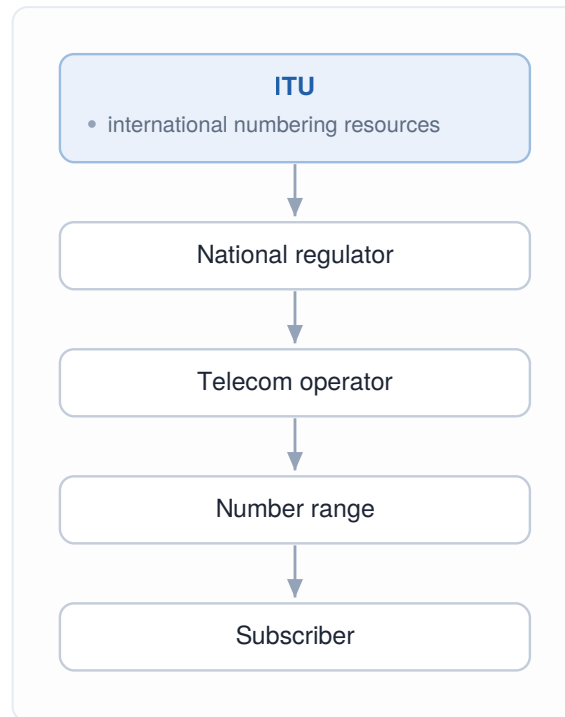


Figure 10.2 ITU → Telecom operator → Subscriber

The exact administrative chain varies by country.

In India, numbering resources are administered within the national telecom regulatory/government framework, while operators receive numbering resources for services.

The number isn't literally "owned" by you in the same sense that you own a physical object.

You have a right to use it under the applicable telecommunications service arrangement.

10.3 What happens when you call another country?

Suppose:

- | | | |
|------------------|---------|-------------------|
| • India | • calls | • +1 212 555 1234 |
| • +91 9876543210 | • USA | |

Conceptually:

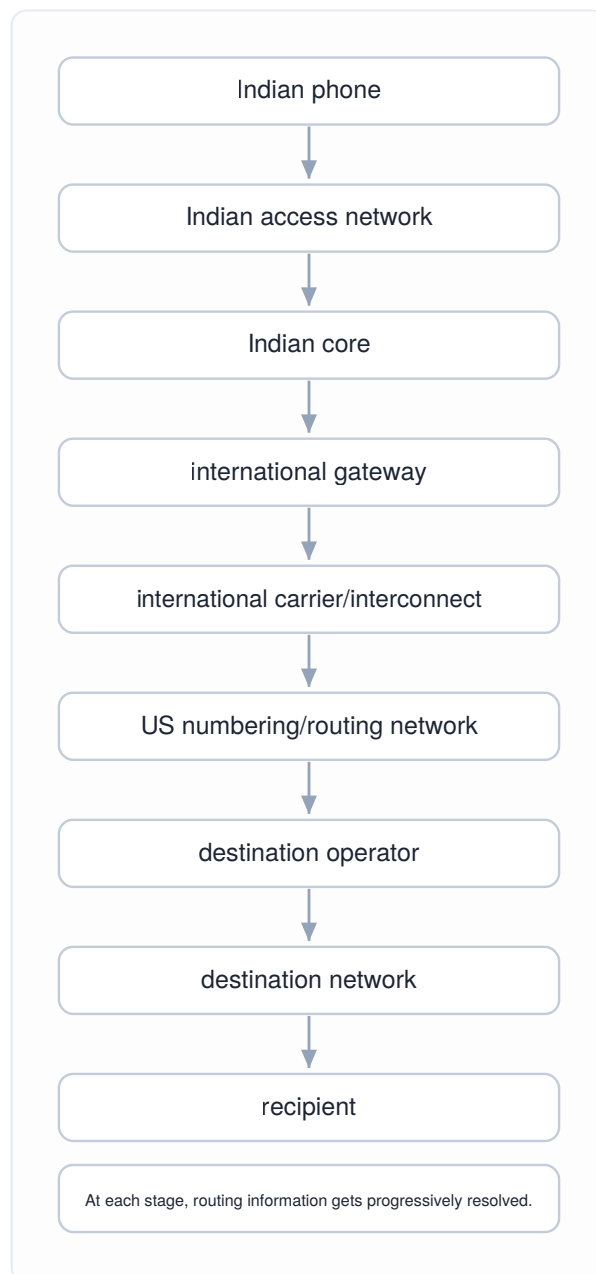


Figure 10.3 Indian phone → US numbering/routing network → At each stage, routing information gets progressively resolved

This is essentially hierarchical distributed routing.

Comparisons and Boundary Cases

11.1 The relationship with DNS

11.2 Phone number vs IP address

11.3 Phone number vs SIM

11.4 One person can have multiple numbers

11.1 The relationship with DNS

There is a fascinating analogy:

DNS

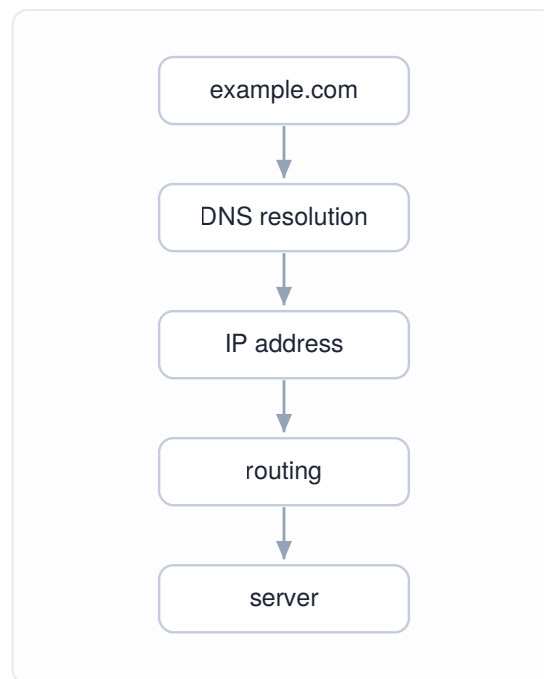


Figure 11.1 example.com → IP address → server

Telephony

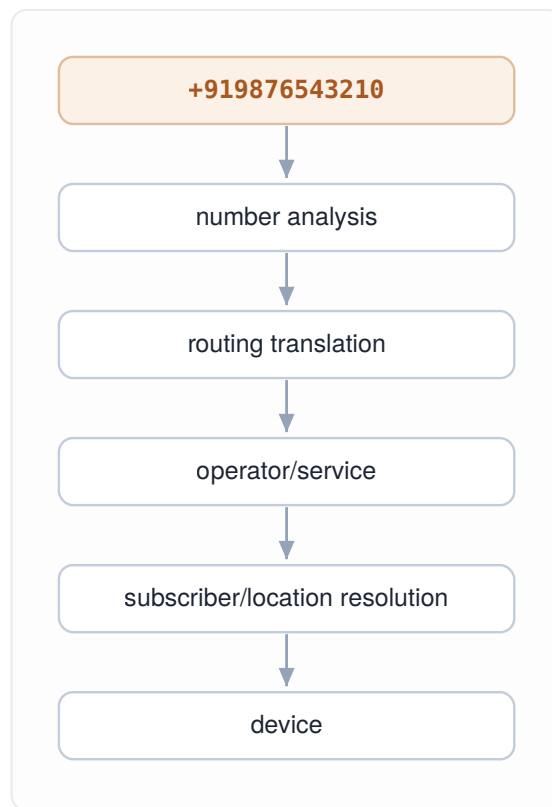


Figure 11.2 +919876543210 → operator/service → device

Both are forms of:

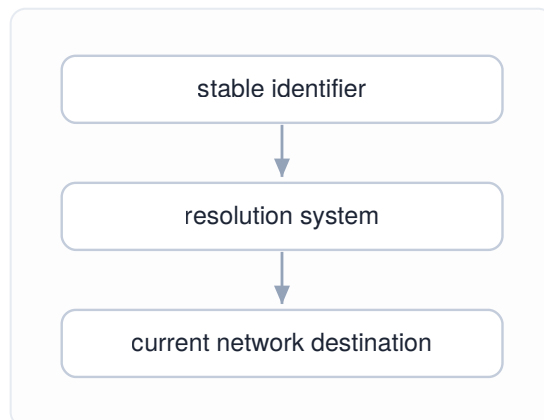


Figure 11.3 stable identifier → current network destination

But telephone numbering predates the Internet by roughly a century.

11.2 Phone number vs IP address

Do not think:

phone number = IP address

They operate at different abstraction layers.

For example:

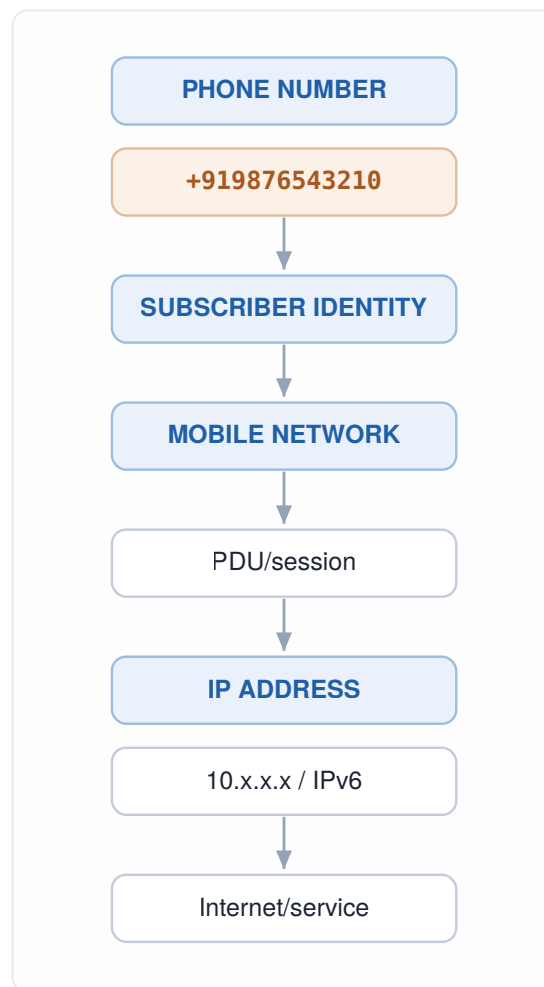


Figure 11.4 Phone number → PDU/session → Internet/service

Your phone can have one telephone number while its packet-network address changes.

11.3 Phone number vs SIM

Likewise:

phone number $\neq$ SIM

You can replace your SIM while retaining your number through provisioning/porting mechanisms.

And:

SIM $\neq$ phone

You can move the subscription between devices.

And:

phone number $\neq$ IMEI

The IMEI identifies equipment.

So a simplified identity graph is:

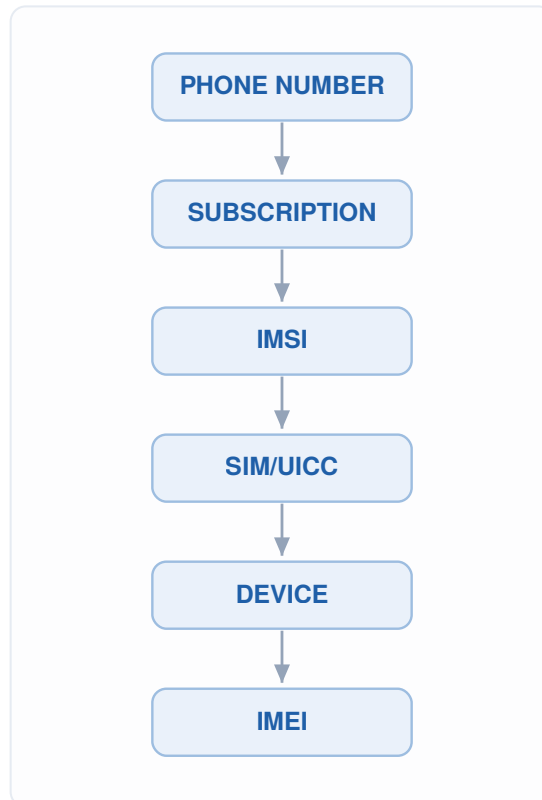


Figure 11.5 Phone number $\rightarrow$ SIM/UICC $\rightarrow$ IMEI

But real-world systems can have multiple numbers, subscriptions and identities associated with one device or account.

11.4 One person can have multiple numbers

There is no fundamental rule:

human = phone number

You can have:

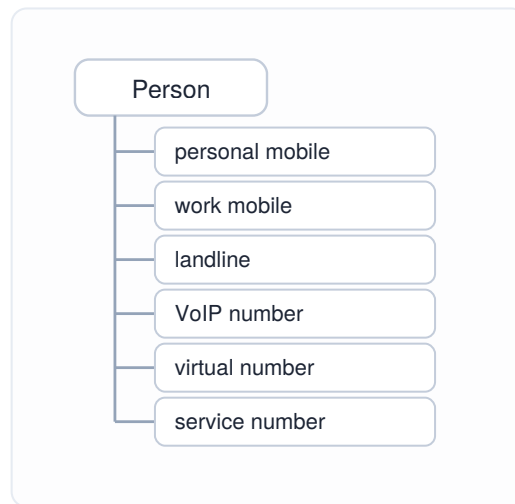


Figure 11.6 Person and its constituent parts

Likewise:

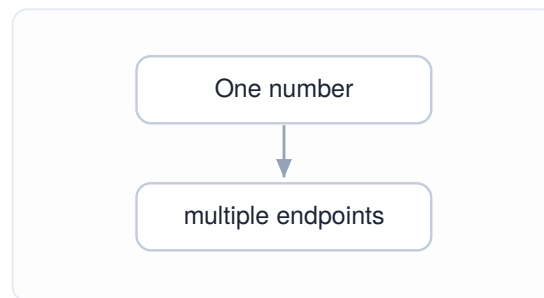


Figure 11.7 One number → multiple endpoints

can exist in business telephony.

For example:

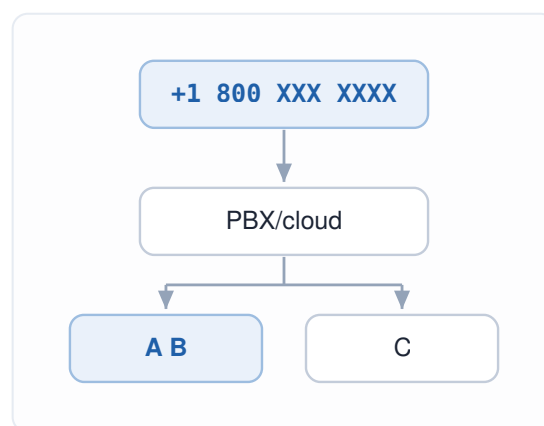


Figure 11.8 +1 800 xxx xxxx → A B / C

So a phone number can identify a service, not necessarily a single physical device.

Special Numbers and Overlaid Services

12.1 Emergency numbers

12.2 Toll-free numbers

12.3 Voicemail

12.4 Caller ID

12.5 SMS uses the same numbering namespace

12.6 WhatsApp demonstrates another layer

12.1 Emergency numbers

Some numbers don't follow normal subscriber numbering.

For example:

112
911
999

These are service identifiers.

The network recognizes them specially:

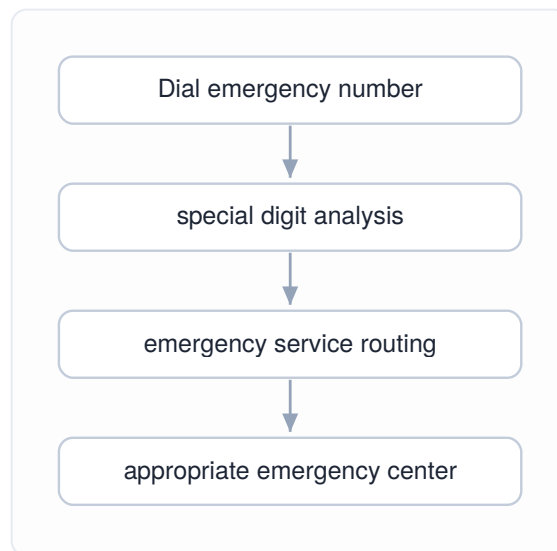


Figure 12.1 Dial emergency number → appropriate emergency center

They therefore demonstrate that the telephone numbering system isn't simply:

number → subscriber

It can be:

number → service

12.2 Toll-free numbers

Similarly:

800...

doesn't necessarily mean:

physical phone at location X

Instead:

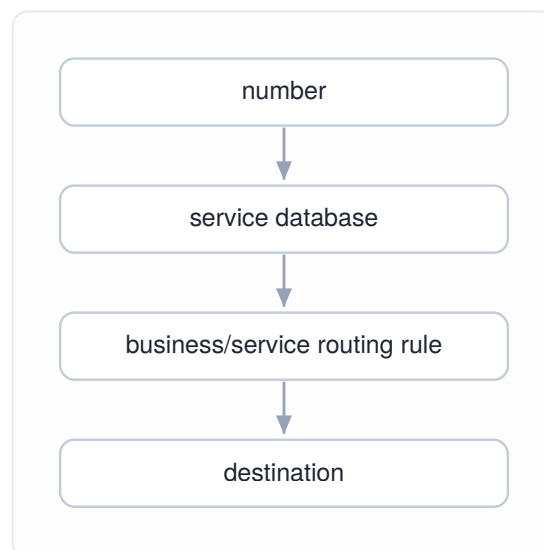


Figure 12.2 number → destination

The network can determine:

- time
- location
- originating carrier
- load
- destination availability

and select an appropriate endpoint.

Telephone numbers therefore evolved into a programmable routing namespace.

12.3 Voicemail

Consider:

+91 9876543210

You call it.

If unavailable:

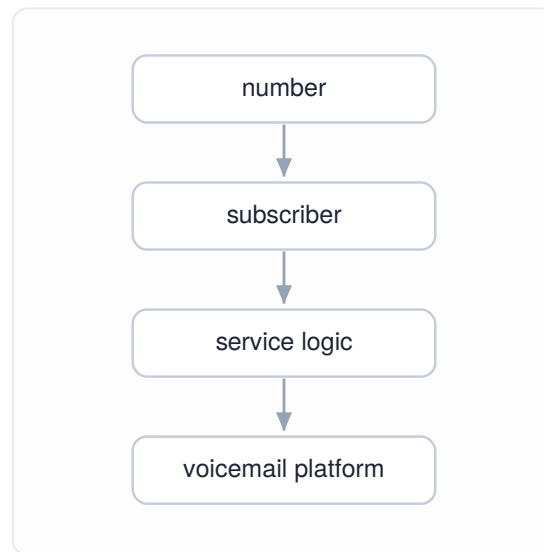


Figure 12.3 number → voicemail platform

The number itself didn't change.

The network/service logic determined:

"this subscriber is unavailable"

and routed the session to another service endpoint.

12.4 Caller ID

When you call someone, their phone can display:

+91 9876543210

That information is transmitted through signaling mechanisms.

It is not the phone magically discovering your number.

Conceptually:

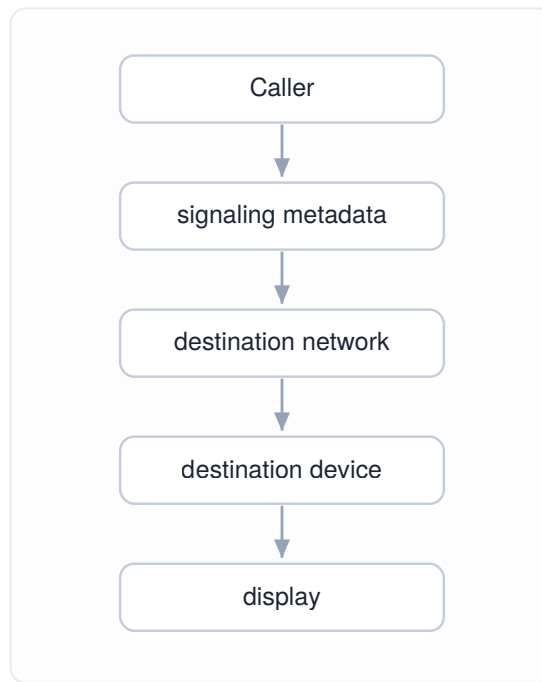


Figure 12.4 Caller → destination network → display

And because signaling information can be manipulated under certain circumstances, caller ID is not cryptographic proof of identity.

That is why spoofing exists.

12.5 SMS uses the same numbering namespace

SMS historically uses telephone numbering as an addressing mechanism.

Simplified:

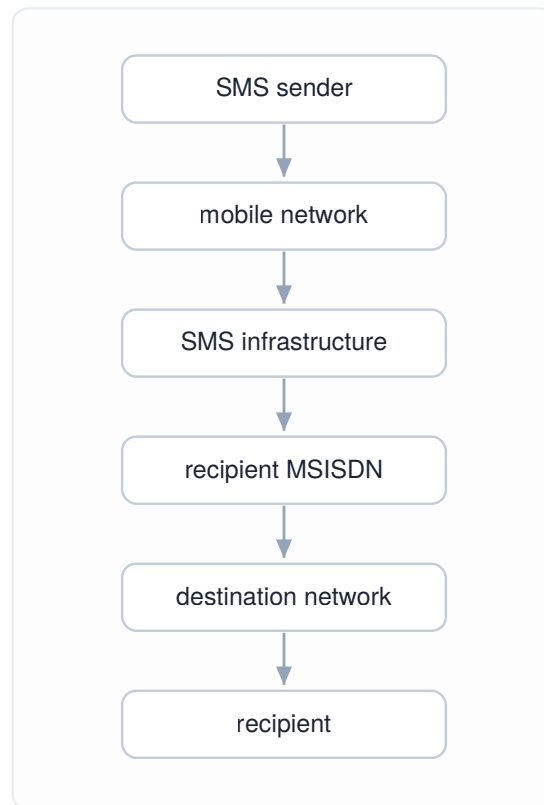


Figure 12.5 SMS sender → recipient MSISDN → recipient

The SMS infrastructure determines where to deliver the message using subscriber/network information.

phone number

became useful beyond voice.

12.6 WhatsApp demonstrates another layer

Suppose WhatsApp registers:

+91 9876543210

WhatsApp uses the telephone number as an account identifier / bootstrap identity.

But after registration, the actual communication doesn't mean:



Figure 12.6 WhatsApp → phone number

Instead:

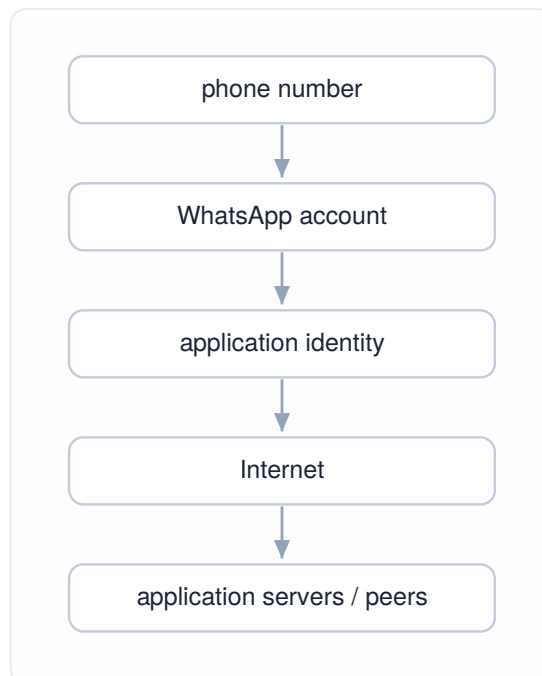


Figure 12.7 phone number → application identity → application servers / peers

This is another example of:

A telephone number can be used as an identifier without the telephone network carrying the actual communication.

The Whole Evolution in One View

13.1 The entire historical evolution

13.2 The most important conceptual transformation

13.3 The modern conceptual model

13.4 The really deep abstraction

13.1 The entire historical evolution

You can compress the history into this:

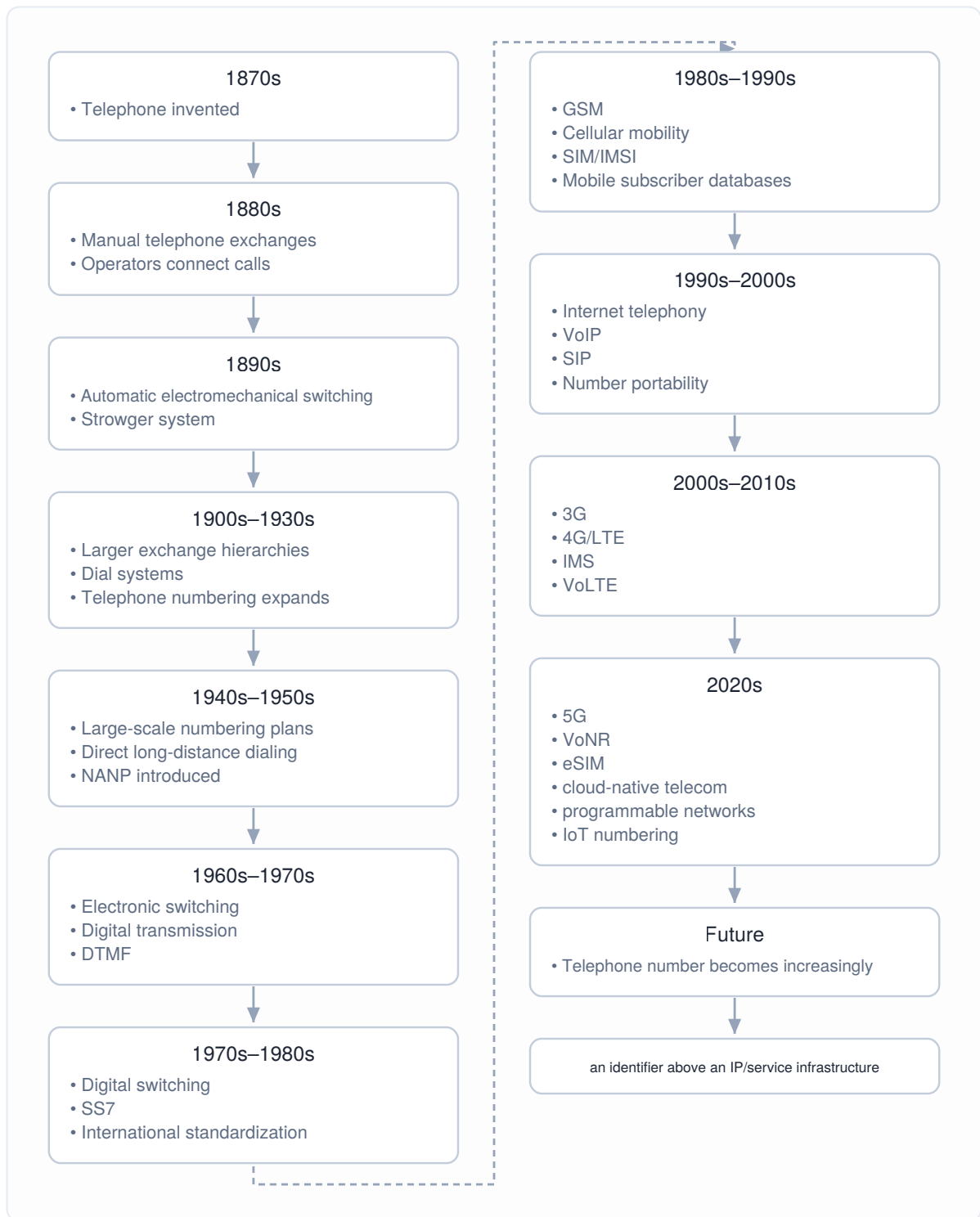


Figure 13.1 1870s → 1970s–1980s → an identifier above an IP/service infrastructure

The key historical pattern is:

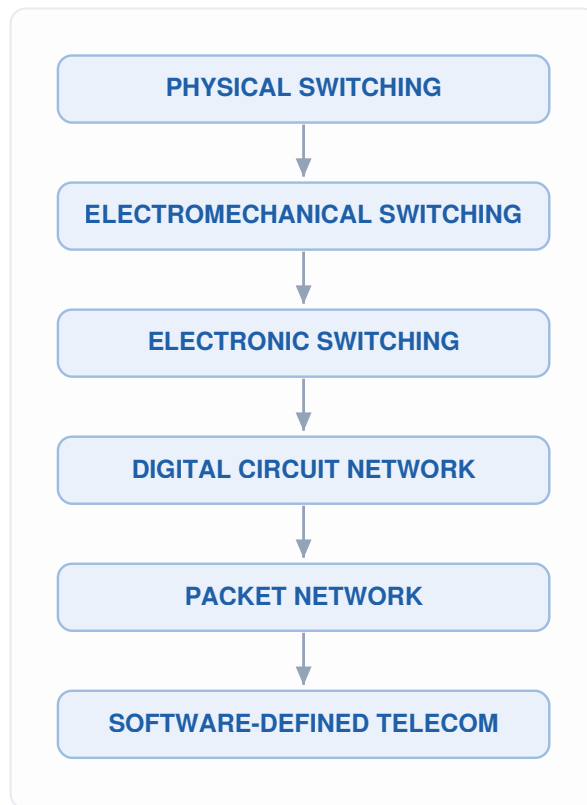


Figure 13.2 Physical switching → Digital circuit network → Software-defined telecom

while:

telephone number

survived almost all of it.

13.2 The most important conceptual transformation

Originally:

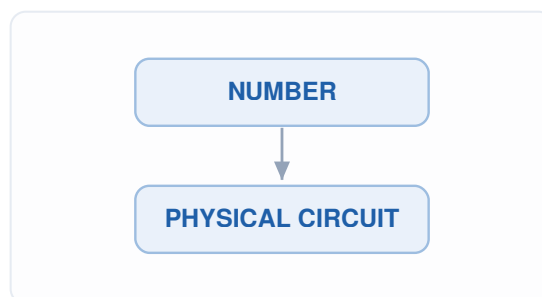


Figure 13.3 NUMBER → Physical circuit

Then:

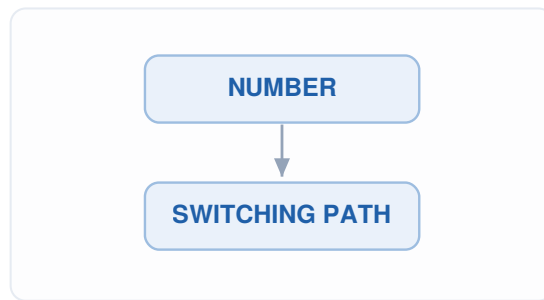


Figure 13.4 NUMBER → Switching path

Then:

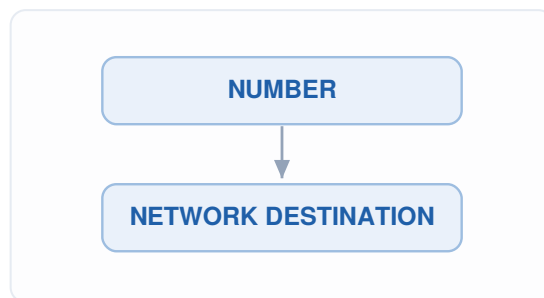


Figure 13.5 NUMBER → Network destination

Then:

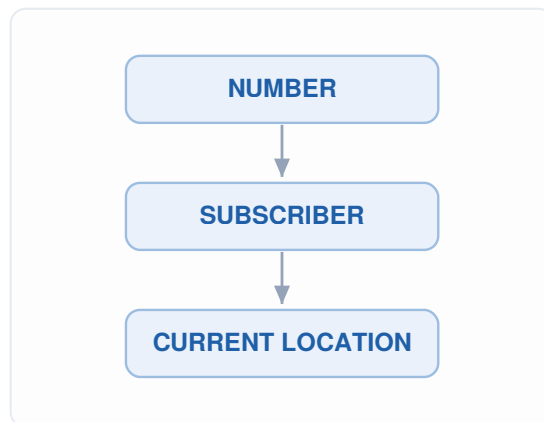


Figure 13.6 NUMBER → Current location

And increasingly:

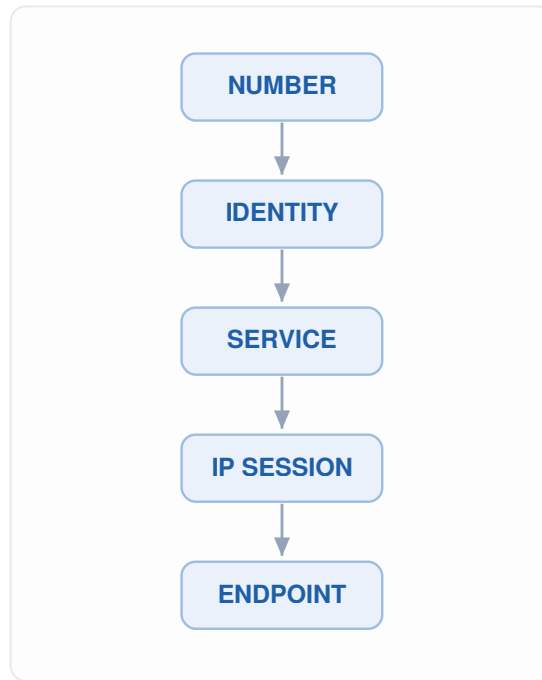


Figure 13.7 NUMBER → SERVICE → ENDPOINT

That is the actual evolution.

13.3 The modern conceptual model

If you want the whole thing represented as one architecture diagram:

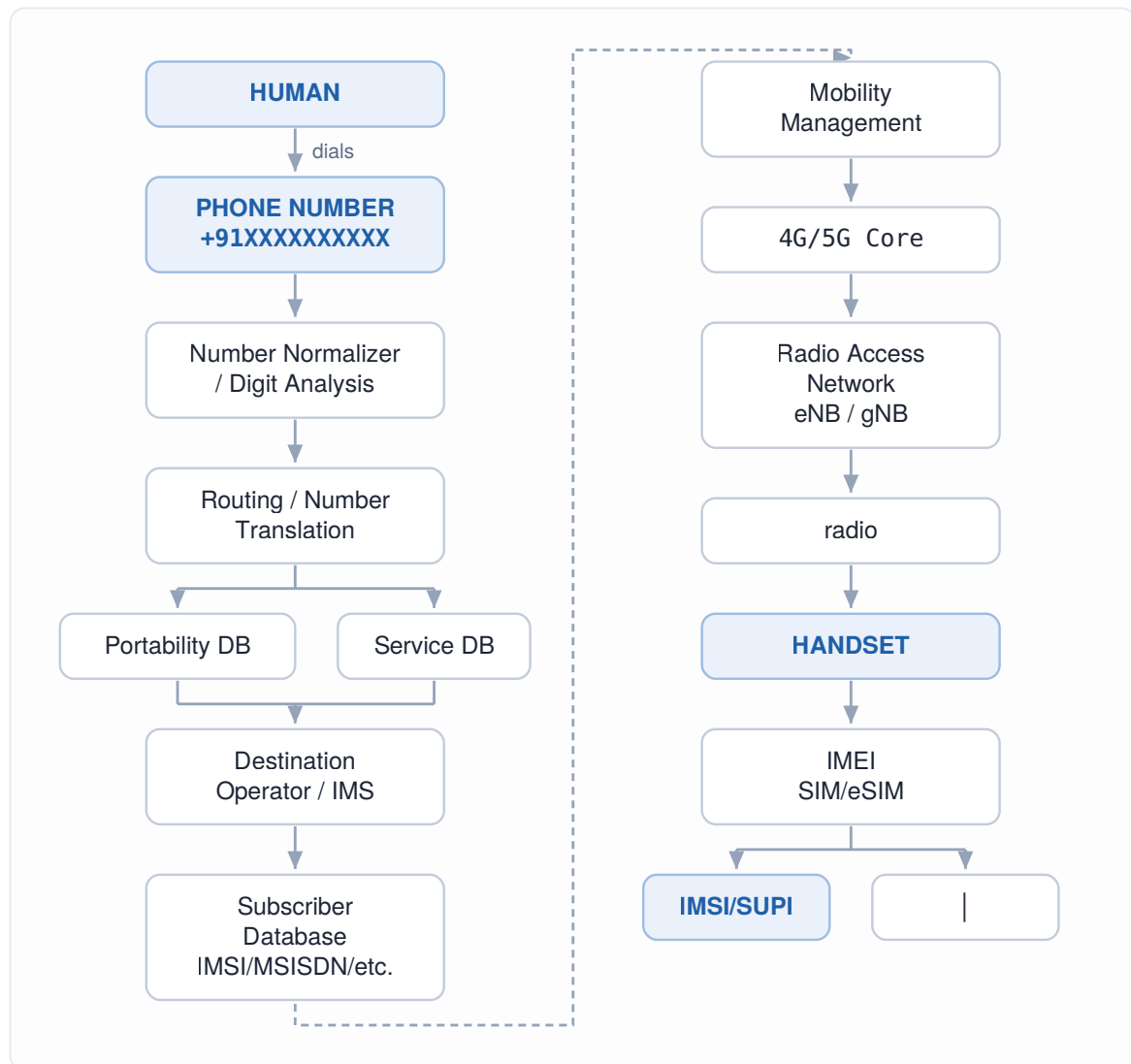


Figure 13.8 HUMAN → Mobility → Imsi/supi / |

And alongside that:

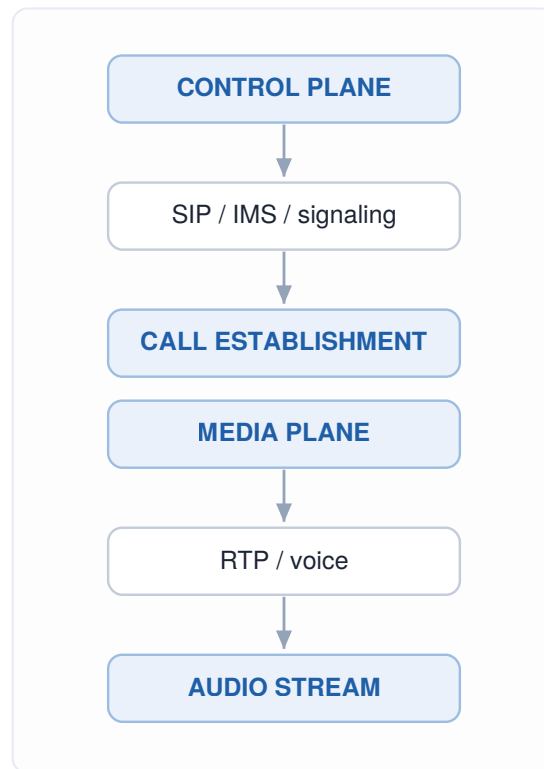


Figure 13.9 Control plane → Media plane → Audio stream

13.4 The really deep abstraction

Since you're studying computer architecture and systems, this is probably the most useful way to think about it.

A telephone number is essentially an indirection layer.

Compare:

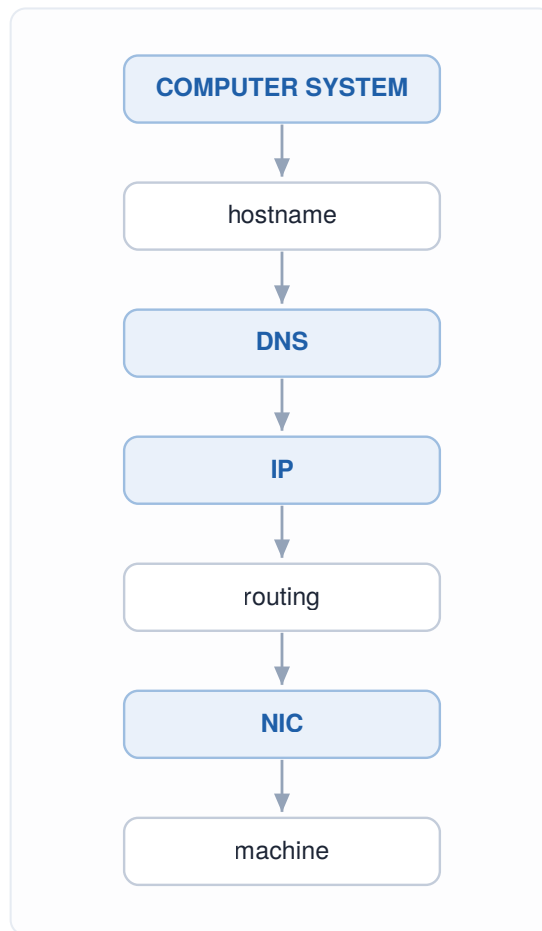


Figure 13.10 Computer system → IP → machine

with:

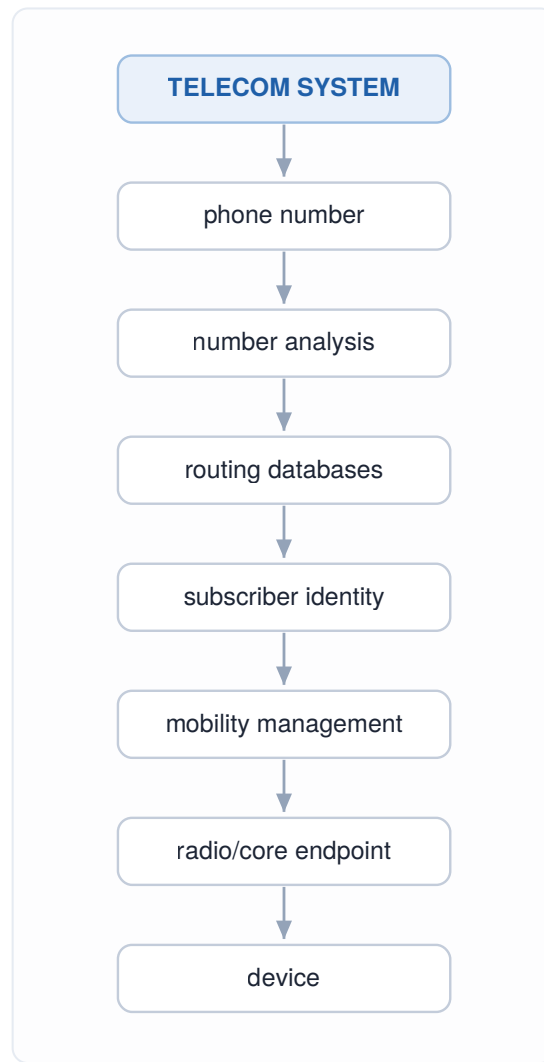


Figure 13.11 Telecom system → subscriber identity → device

The telephone network spent more than a century gradually moving the mapping:

number → physical circuit

toward:

number → logical identity → dynamic endpoint

That is why the number itself has remained remarkably stable even though practically every layer underneath it has been replaced.

And that is also why you should not memorize:

"The first four digits tell you the operator."

That is an artifact of a particular numbering/allocation regime, not the fundamental architecture. Number portability exists specifically because the logical identifier and underlying network destination are separable.

The hierarchy to remember

If you want one mental model that survives almost every telecommunication technology:

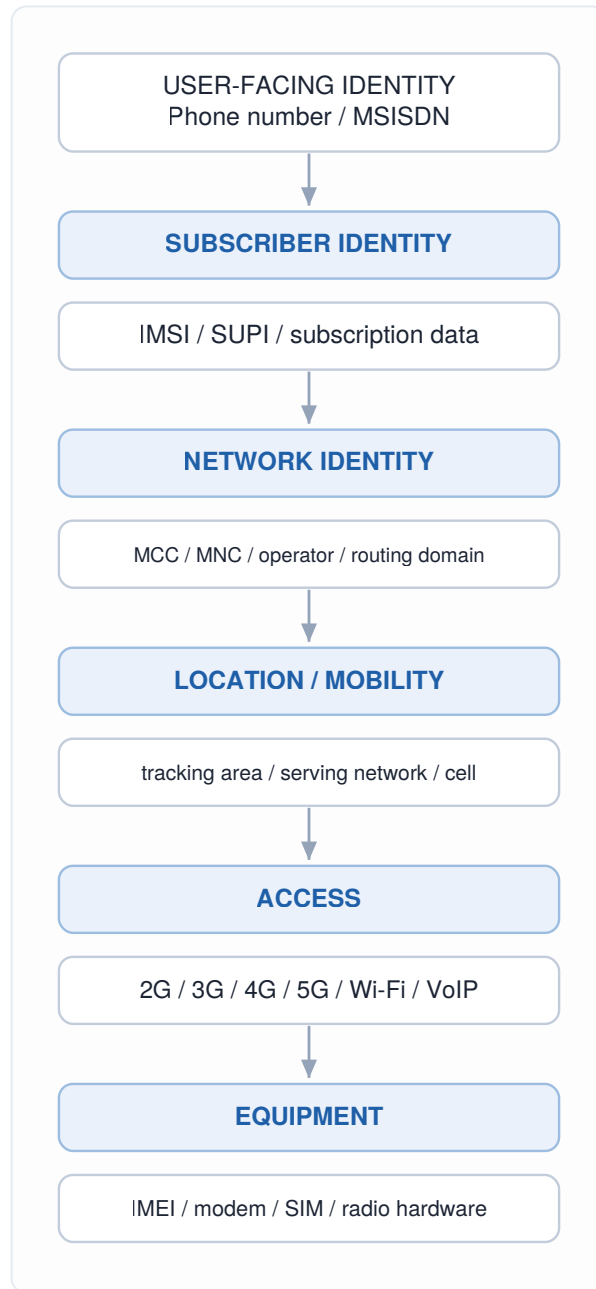


Figure 13.12 User-facing identity → Location / mobility → IMEI / modem / SIM / radio hardware

That's the architecture underneath the innocent little 10 digits humans type into a contact list.

The standards themselves are still evolving: the ITU issued the current E.164 revision in February 2026, including numbering provisions for newer categories such as IoT/M2M. ^{ITU}

Primary technical references

Primary technical references

- ITU-T E.164, International Public Telecommunication Numbering Plan
- ITU-T E.164.1, allocation and reclamation of country codes
- NANPA, history and structure of the North American Numbering Plan
- TRAI, Mobile Number Portability framework
- ETSI/3GPP TS 23.008, mobile subscriber data architecture
- GSMA mobile identity glossary

The natural next layer to study is exactly what happens inside the network after you press the Call button, down to MSISDN → IMSI/SUPI → HLR/HSS/UDM → MSC/IMS → SS7/Diameter/SIP → gNB → paging → bearer establishment → RTP voice. That is where the phone-number abstraction finally collides with actual telecom machinery.

Glossary of Terms

Every term below appears somewhere in this volume. Expansions are given where an acronym has one; the definitions describe the sense in which the term is used here rather than every sense it carries in the wider literature.

0–9

5G-GUTI

Globally Unique Temporary Identity

A temporary identity allocated by the AMF and used in place of the SUPI for most signalling, so the permanent identity is rarely broadcast.

A

AuC

Authentication Centre

The GSM-era node holding the secret keys used to authenticate SIMs.

B

BSC

Base Station Controller

The GSM node controlling a group of base stations, handling radio resources and handovers between them.

BTS

Base Transceiver Station

The GSM radio site — the equipment that actually transmits and receives over the air.

C

Caller ID

Calling-party information delivered as signalling metadata alongside the call, not derived from the audio path.

Control plane

The signalling that decides what should happen: who is calling whom, whether they are authorised, where the destination is, what resources are needed.

CSCF

Call Session Control Function

The family of SIP servers at the heart of IMS: the proxy (P-CSCF), interrogating (I-CSCF) and serving (S-CSCF) roles.

D

DNS

Domain Name System

The internet's name-resolution system, used in this book as the closest analogy to telephone number analysis and routing databases.

DS0

A single 64 kbit/s digital voice channel — 8 kHz sampling at 8 bits per sample — the basic unit multiplexed onto TDM trunks.

DTMF

Dual-Tone Multi-Frequency

Touch-tone signalling: each digit is sent as a pair of audio tones, replacing the pulse counting of rotary dials.

E

E.164*ITU-T Recommendation E.164*

The international public telecommunication numbering plan. Defines the structure of international numbers as a country code followed by a national significant number, historically within a 15-digit maximum.

eNB

The LTE base station, the 4G counterpart of the gNB.

eSIM

A SIM implemented as a reprogrammable embedded module, allowing a subscription to be provisioned without swapping a physical card.

ETSI*European Telecommunications Standards Institute*

European standards body, a founding partner of 3GPP.

Exchange

A telephone switch serving a group of subscriber lines; historically the node that interpreted dialled digits as switching instructions.

F

FCC*Federal Communications Commission*

The United States communications regulator.

G

gNB

The 5G base station. It terminates the radio protocol stack and connects to the core over N2 (control) and N3 (user plane).

GSM*Global System for Mobile Communications*

The second-generation digital cellular standard that introduced the SIM, the IMSI and the HLR/VLR mobility model.

GSMA*GSM Association*

The industry body representing mobile operators; maintains a number of mobile identity and interconnect specifications.

H

Handover

Moving an active connection from one cell to another without dropping the session.

HLR*Home Location Register*

The master subscriber database of a GSM network, holding subscription data and a pointer to where the subscriber is currently registered.

HSS*Home Subscriber Server*

The IMS/LTE subscriber database holding authentication data, service profiles and registration state.

I

ICCID*Integrated Circuit Card Identifier*

The serial number of the physical SIM card itself, distinct from the subscriber identity stored on it.

IMEI*International Mobile Equipment Identity*

Identifies the handset hardware rather than the subscriber. Moving a SIM to another phone changes the IMEI while the IMSI and MSISDN stay the same.

IMS*IP Multimedia Subsystem*

The standardised SIP-based service core that operators use to deliver voice and other multimedia over their packet networks, and to interwork with the PSTN.

IMSI*International Mobile Subscriber Identity*

The permanent identity of a subscription, stored on the SIM. Composed of MCC, MNC and MSIN. Normally never revealed to the caller and increasingly replaced on the air interface by temporary identities.

IoT / M2M*Internet of Things / machine-to-machine*

Categories of non-human subscriptions for which dedicated numbering ranges have been defined in recent revisions of E.164.

IP*Internet Protocol*

The packet-addressing and routing protocol. A device's IP address changes with attachment; the telephone number does not.

ISDN*Integrated Services Digital Network*

A digital circuit-switched successor to analogue subscriber lines, carrying voice and data over the same access.

ITU*International Telecommunication Union*

The United Nations agency that allocates international numbering resources and publishes the E-series numbering recommendations.

ITU-T*ITU Telecommunication Standardization Sector*

The branch of the ITU that issues telecommunication standards, including E.164.

IVR*Interactive Voice Response*

An automated system that answers a call and routes or serves the caller by menu.

M**MAC***Medium Access Control*

The radio layer that schedules transmissions, multiplexes logical channels and runs HARQ re-transmissions. Unrelated to an Ethernet MAC address.

MCC*Mobile Country Code*

The leading field of an IMSI identifying the subscriber's home country.

MNC*Mobile Network Code*

The field following the MCC that identifies the home operator within that country.

MNP*Mobile Number Portability*

The arrangement that lets a subscriber keep a telephone number when changing operator, breaking the old assumption that a number prefix identifies the serving network.

MSC*Mobile Switching Centre*

The circuit-switched call-control and switching node of GSM-era networks. It has no single successor in 5G: its duties are split across the AMF, SMF, UPF and IMS.

MSIN*Mobile Subscriber Identification Number*

The operator-assigned portion of an IMSI that identifies the individual subscription.

MSISDN*Mobile Station International Subscriber Directory Number*

The dialable telephone number attached to a mobile subscription — the number the outside world knows. It is a service identifier, not a device address.

N**NANP***North American Numbering Plan*

The shared numbering plan for the United States, Canada and several other territories, reached with country code +1 and structured as NPA-NXX-XXXX.

NANPA*North American Numbering Plan Administrator*

The body that administers number resources within the NANP.

NPA*Numbering Plan Area*

The area code in the NANP — the first three digits of a ten-digit number.

NSN*National Significant Number*

The part of an international number that follows the country code.

Number portability

The ability to keep a telephone number when changing operator. It severs the historical link between a number prefix and a serving network, and forces a database lookup into the routing path.

NXX

The central-office or exchange prefix in a NANP number: the second group of three digits.

P

Paging

The procedure by which the network locates an idle device. It is broadcast only within the tracking area where the device was last known, not across the whole network.

PBX

Private Branch Exchange

A private switch inside an organisation, allowing one external number to reach many internal endpoints.

PCM

Pulse-Code Modulation

The sampling-and-quantisation scheme that turns an analogue voice signal into the classic 64 kbit/s digital stream.

PDU session

Protocol Data Unit session

The logical connection between the device and the UPF that provides IP connectivity. Voice typically uses a session dedicated to IMS.

PSTN

Public Switched Telephone Network

The traditional circuit-switched telephone network. Modern IP telephony still interworks with it through gateways.

R

RAN

Radio Access Network

The radio side of a mobile network: the base stations and the protocols between them and the device.

Resource block

The smallest unit of radio spectrum and time the scheduler can allocate to a device.

Roaming

Using a visited operator's network while subscription data and authentication remain anchored at the home network.

RTP

Real-time Transport Protocol

The protocol (RFC 3550) that carries the actual media. Its header supplies a sequence number, a timestamp and an SSRC so the receiver can reorder, time and separate streams.

S

SIM / UICC

Subscriber Identity Module

The secure element holding the subscription identity and the keys used to authenticate it. The card is the UICC; the SIM is the application running on it.

SIP

Session Initiation Protocol

The text-based signalling protocol (RFC 3261) used to create, modify and terminate multimedia sessions. It sets up the call but carries none of the audio.

SS7

Signalling System No. 7

The out-of-band signalling network of the digital PSTN and of 2G/3G mobile networks. It separates call control from the voice path and carries database queries such as roaming and portability lookups.

Strowger switch

The first widely deployed automatic telephone exchange, in which dial pulses drove electromechanical selectors directly — the origin of the number-as-instruction model.

SUPI

Subscription Permanent Identifier

The 5G equivalent of the IMSI. In 5G it is not sent in the clear over the radio; a concealed form is used during initial registration.

T

TDM

Time-Division Multiplexing

Carrying many fixed-rate channels on one trunk by giving each a repeating time slot.

Toll-free number

A number that resolves through a service database to a destination chosen by rule rather than mapping to any fixed line.

Tracking area

A group of cells the network treats as one location unit. The device reports when it moves between tracking areas, which bounds the scope of paging.

TRAI

Telecom Regulatory Authority of India

India's telecommunications regulator, responsible among other things for the mobile number portability framework.

Trunk

A shared high-capacity link between exchanges, as distinct from the individual line to a subscriber.

U

UDM

Unified Data Management

The 5G subscriber data function — the successor in role to the HLR and HSS.

UE

User Equipment

The 3GPP term for the subscriber's device — handset plus SIM/USIM.

V

VLR

Visitor Location Register

A local database holding temporary records for subscribers currently attached to a given serving area.

VoIP

Voice over IP

Carrying telephone speech as packets over an IP network rather than over a dedicated circuit.

VoLTE

Voice over LTE

Voice carried as IMS-controlled packets over a 4G LTE radio and core network.

VoNR

Voice over New Radio

The 5G Standalone equivalent: IMS voice carried over 5G NR with a 5G core.